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基於鯰魚式的深度強化學習之射頻鏈與反射元件之節
能聯合開關控制在 RIS 輔助 6G 網路

CDRL: Catfish-Effect-Based Deep Reinforcement
Learning for Energy-Efficient Joint Switching of RF
Chains and Reflecting Elements in RIS-Aided 6G
Networks

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論文提要內容：

可重構智慧表面 (RIS) 是一種新興的通訊技術，透過反射電磁訊號來增強鏈路性能。為因應 6G 毫米波下行 MU-MISO 系統中主/被動 RIS 帶來的高功耗與高維離散-連續混合控制難題。本文提出一套鯰魚式深度強化學習架構 (Catfish-effect-based DRL, 簡稱 CDRL)，用以同時最佳化 BS 與 RIS 的元件開關控制及波束成形，最終達到最大化能源效率的目的。本文提出的 CDRL 是由 catfish 與 main 雙代理組成。Catfish agent 透過能效門檻分流專注蒐集高獎勵經驗，並配以較大折扣係數及競爭獎勵機制。Main agent 則處理一般情境，並在隨機干預階段 catfish agent 透過將自身的長期知識注入 main agent，實現經驗級知識轉移與樣本效率提升。此「雙代理-雙回放」設計為將 DRL 應用於主動 RIS 結合元件開關之能源效率最佳化研究，並有效避免單代理策略僵化。模擬結果顯示，CDRL 於主/被動 RIS 場景的能源效率均優於基礎 DRL、數學解析法 BPOF 及 Teacher-Student 框架。能在滿足速率約束下顯著降低總功耗，驗證了鯰魚效應結合 DRL 於無線資源配置問題的可行性，並為 6G 綠色通訊提供新方向。這項研究的動機在於透過基地台控制技術與主動智能反射面間的協同作用，在維持一定的傳輸品質下，實現 6G 綠色通訊的節能和省電的目標。

關鍵字：能源效率、混合波束成型、可重構智能表面、深度學習。

ABSTRACT

CDRL: Catfish-Effect-Based Deep Reinforcement Learning for Energy-Efficient Joint Switching of RF Chains and Reflecting Elements in RIS-Aided 6G Networks

by
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DEPARTMENT: COMPUTER SCIENCE & INFORMATION ENGINEERING

MAJOR: HYBRID BEAMFORMING AND PHASE SHIFT OPTIMIZATION WITH ON-OFF CONTROL OF RF CHAIN AND REFLECTING ELEMENT FOR ENERGY-EFFICIENCY RIS-AID DOWNLINK COMMUNICATIONS

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A reconfigurable intelligent surface (RIS) is an emerging communications technology that enhances link performance by reflecting electromagnetic signals. To address the challenges of high-power consumption and hybrid discrete-continuous control introduced by active/passive RIS in millimeter-wave downlink MU-MISO systems. This thesis proposes a Catfish-effect-based deep reinforcement learning framework (Catfish-effect-based DRL, CDRL) for simultaneously optimizing BS and RIS element-switch control and beamforming, ultimately maximizing energy efficiency (EE). In this thesis, the CDRL framework comprises two agents: the catfish agent and the main agent. The catfish agent diverts according to an energy-efficiency threshold to focus on collecting high-reward experiences, and is paired with a larger discount factor and a competitive reward mechanism. The main agent handles general scenarios and, during the random intervention phase, the catfish agent injects its long-term knowledge into the main agent, achieving experience-level knowledge transfer and improved sample efficiency. This dual-agent, dual-replay design advances DRL-based EE optimization for RIS systems with joint switching control, and effectively mitigates policy stagnation issues often observed in single-agent settings. Simulation results demonstrate that CDRL outperforms baseline DRL, the analytical method BPOF, and Teacher-Student frameworks in both active and passive RIS scenarios. CDRL achieves significant power savings under rate constraints, validating the feasibility of integrating the catfish effect with DRL for wireless resource allocation. This thesis highlights a new direction for green communications in 6G, motivated by the synergy between BS control strategies and active RISs to reduce energy consumption while maintaining transmission quality.

Key words: Energy efficiency, hybrid beamforming, reconfigurable intelligent surface, Deep Learning.

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Chapter 1

Introduction

To accommodate the massive transmission demands of 6G, millimeter-wave (mmWave) massive multiple-input single-output/multiple-input multiple-output (Massive MISO/MIMO) technology has become a popular research direction and practical strategy in this field [4]-[5][7]-[14][16]-[17][19]-[20]. mmWave transmission enables increased spectrum utilization and enhanced data rates. However, due to its high-frequency characteristics, it results in relatively short propagation distances and severe path loss [10, 14]. MIMO technology effectively addresses these issues by supporting multiple users, improving signal strength, and extending communication coverage. MIMO allows the multiple antennas at the base station to perform beamforming, which significantly enhances the performance of wireless systems by enabling directional signal transmission. This improves the signal-to-noise ratio (SNR), mitigates co-channel interference, and increases network capacity [1]. Conventionally, MIMO precoding is realized through a fully digital (FD) method, where each antenna element is equipped with a dedicated radio frequency (RF) chain [23]. However, deploying an RF chain for every antenna at the base station leads to high hardware cost and power consumption [13, 14]. Therefore, hybrid beamforming (HBF) is commonly adopted under this architecture to achieve satisfactory beamforming gains with reasonable hardware complexity and power consumption [10, 13, 14].

Due to the susceptibility of millimeter-wave (mmWave) communications to environmental blockages such as walls and trees, which cause severe penetration loss, reconfigurable intelligent surfaces (RIS) have been proposed to assist hybrid beamforming (HBF) [14]. RIS is an

emerging next-generation technology that controls the reflection of electromagnetic waves by reconfiguring the electromagnetic properties of the surface. RIS can effectively improve signal transmission efficiency [5, 10, 14], expand wireless network coverage, and enhance transmission quality [4, 16, 18]. These reflective surfaces are strategically deployed between the transmitter and receiver, providing users with an additional communication link and improving signal strength [10]. However, this induces multiplicative fading, leading to limited capacity gains for conventional RIS [9]. Specifically, the communication channel reflected via RIS experiences two stages of propagation loss, meaning that the equivalent path loss of the transmitter-RIS-receiver link is the product of the transmitter-to-RIS and RIS-to-receiver path losses [10]. In general, conventional RIS consists of a large number of passive elements, each capable of independently adjusting the phase of the incident electromagnetic wave to redirect the signal toward the target receiver [18]. In contrast, active RIS is a new type of RIS that can simultaneously adjust the phase shift and amplify the received signal, thereby mitigating signal attenuation and eliminating the double fading effect associated with multiplicative fading [9]. This enables active RIS to provide better communication performance under weak signal power conditions.

Network energy saving is critical for environmental sustainability, reducing environmental impact (e.g., greenhouse gas emissions), and lowering operational costs [2]. In wireless communications, although the use of beamforming techniques at the transmitter via MIMO and RIS-assisted transmission brings several advantages as mentioned above, such considerations often lead to high power consumption. Therefore, improving energy efficiency while maintaining acceptable transmission performance has become a vital issue. Laselva *et al.* evaluated the potential of each radio domain, where radio adaptation in the antenna domain involves adjusting the number of antennas (or antenna ports) used for transmission by the base transceiver station (BTS), which in turn affects the total transmit power [3]. This implies that deactivating specific antenna elements can reduce energy consumption, and this concept can also be extended to RIS. Since RIS requires power supply to enable effective signal re-

flexion, selectively activating and deactivating reflecting elements can optimize RIS energy usage [9]. However, due to the large number of elements involved and the mutual interference among multiple users and multiple RIS units, jointly adjusting the phase shifts and on-off states becomes a complex challenge. Therefore, the proposed solution adopted in this thesis will be presented in the following chapters.

With the increasing structural and functional complexity of RIS, efficiently controlling a large number of RIS elements—including phase adjustment, on-off switching, and even power amplification—has become a major challenge, especially under multi-user scenarios, dynamic channels, or imperfect channel state information [12, 17, 19, 20]. Traditional optimization approaches, such as convex programming and heuristic methods, often struggle to balance efficiency and accuracy in these high-dimensional and non-convex problems [11, 12, 17]. As a result, machine learning (ML) is increasingly regarded as a promising tool for assisting resource allocation in communication systems, particularly due to its ability to model unknown environments and complex dependencies. Among various ML techniques, deep reinforcement learning (DRL) exhibits strong advantages in handling high-dimensional and sequential decision-making problems [19, 20]. Compared with traditional methods, DRL is especially beneficial in wireless systems with time-varying radio channels. By modeling the communication system as a Markov Decision Process (MDP), DRL can solve complex optimization problems without explicitly formulating system models such as channel or mobility models, instead learning through interaction with the environment and observing reward signals [11, 15, 17, 20]. DRL is particularly well-suited for hybrid action spaces involving both discrete actions (e.g., RIS switching) and continuous actions (e.g., power amplification) [17, 19, 20]. These features make DRL one of the most active research directions for intelligent RIS control.

To address these issues, we propose a novel Catfish-effect-based deep reinforcement learning algorithm (CDRL), as illustrated in Fig. 1. We design two task-oriented agents, where the Catfish Agent provides differentiated experiences to effectively stimulate the Main Agent to explore better policies, thereby avoiding local optima. In summary, the contributions of this

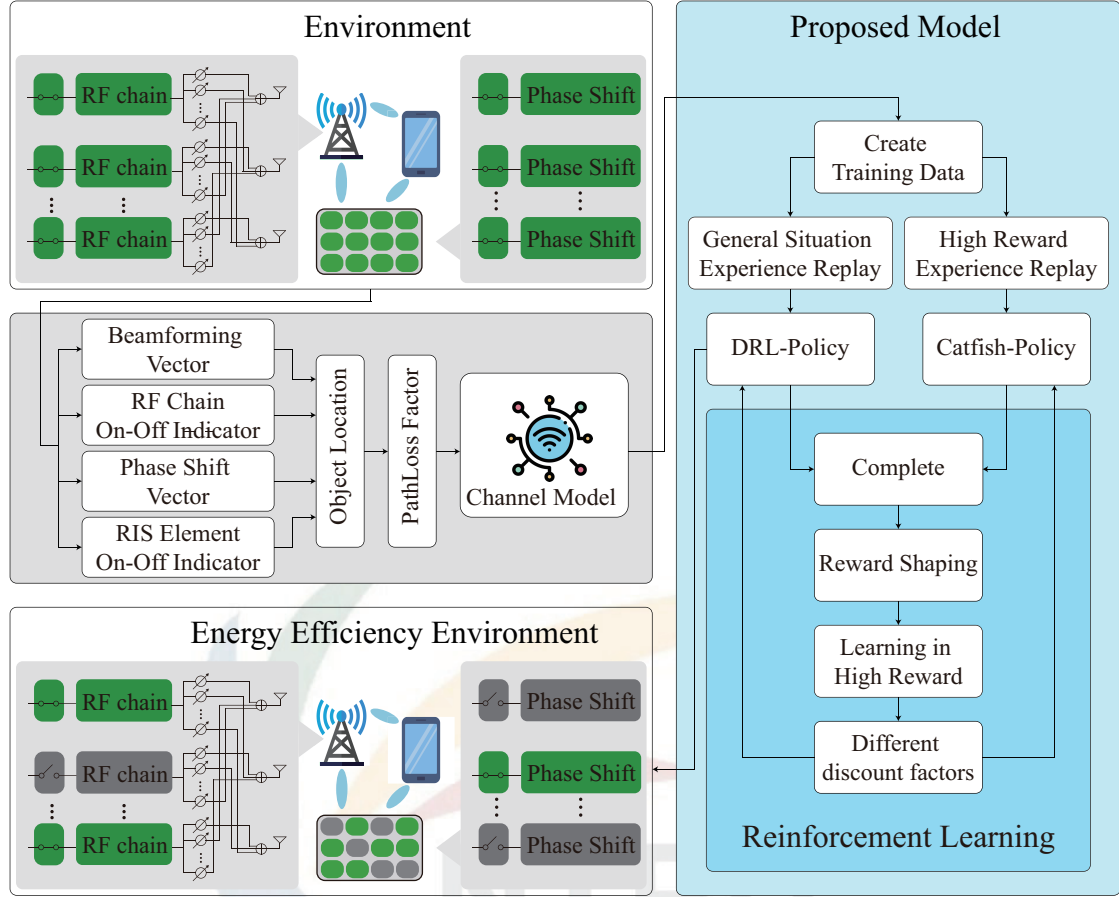


Figure 1.1: Our proposed CDRL scheme.

thesis are outlined as follows:

- To the best of our knowledge, the proposed Catfish-effect-based deep reinforcement learning (CDRL) framework is the first to address the integration of active RIS with element switching control using DRL. CDRL employs two competing DRL agents to optimize communication performance.
- In the proposed CDRL approach, we design three core mechanisms: all experiences are separated based on an energy efficiency (EE) threshold, the two DRL agents are assigned different discount factors to focus on distinct objectives, and the catfish agent is provided with an adaptive competitive reward mechanism to promote mutual competition between the agents.

- In this study, we evaluate our method against a baseline DRL approach, an analytical solution under the same communication environment, and a teacher-student framework. Experimental results demonstrate that our method outperforms both the mathematical solution and teacher-student models under the same conditions, validating the superiority of our approach.

The rest of the thesis is organized as follows. Chapter 2 describes related work and motivation. Chapter 3 introduces the system model, problem formulation, basic concepts, and comparison of Catfish-DRL with adversarial models. Chapter 4 describes the CDRL scheme. Chapter 5 shows the simulation results. Finally, chapter 6 concludes the thesis.



Chapter 2

Research Background

This section 2.1 describes related work, and section 2.2 discusses research motivation.

2.1 Related works

Previous research on network transmission rate or energy efficiency has often relied on mathematical optimization. Under the passive RIS setting, Dejonghe *et al.* [4] proposed a practical, low-cost, and high-performance RIS system design based on semi-definite programming (SDP) to achieve RIS beamforming optimization. In the domain of artificial intelligence-based approaches, Huang *et al.* [11] investigated the joint design of BS transmit beamforming and RIS phase shifts to maximize the sum rate of a multi-user downlink MISO system using DRL, successfully opening a new research direction for deep reinforcement learning. As a result, a growing number of studies have explored deep reinforcement learning and other AI-based solutions. For example, Saglam *et al.* [12] proposed a novel DRL-based modification for RIS-assisted systems, highlighting the adaptability and flexibility of deep reinforcement learning under hardware impairments and limited channel state information. Zhou *et al.* [15] introduced an innovative Co-HDRL algorithm for intelligent hierarchical decision-making in wireless networks to address problems characterized by different time scales.

The multiplicative fading effect limits the capacity gains of passive RIS in wireless communication environments, leading to increasing research on active RIS architectures [6–8]. Active RIS can actively reflect and amplify signals, functioning similarly to relay base stations. Fo-

tock *et al.* [6] considered maximizing energy efficiency (EE) in interference-limited RIS-assisted wireless networks and compared the performance of active and near-passive RIS. Their developed algorithm jointly optimizes RIS reflection coefficients, mobile user transmit power, and the base station's linear receive filters. Wang *et al.* [7] applied an alternating optimization (AO) algorithm based on majorization-minimization (MM) and fractional programming (FP) methods to solve the problem. The objective is to maximize the achievable downlink rate under the transmit power constraints of the base station (BS) and the dual active RISs. de Souza *et al.* [8] proposed an EE optimization framework for active RIS-assisted massive MIMO (mMIMO) systems and designed a low-complexity RIS configuration algorithm. They reformulated the problem as an equivalent quadratically constrained quadratic program (QCQP), from which novel closed-form solutions for the phase and amplitude of each active RIS element were derived to maximize EE. These results demonstrate that active RIS positively contributes to enhancing energy efficiency.

However, a high count of active elements drives up the system's total energy draw, thereby lowering energy efficiency (EE). As a result, growing research has focused on element switching control [19][20]. Under the passive RIS setting, Faisal *et al.* [19] focused on optimizing the states of quantized phase shifters and RIS elements using an efficient DRL-based approach. The agent determines which RIS elements to activate or deactivate based on dynamic information from cellular networks under varying channel state information and total rate requirements. Cao *et al.* [20] studied ON/OFF-based reflection and binary phase-shift keying modulation schemes, developing a new RIS-assisted MISO SR system and a DRL-based framework to solve the EE maximization problem. In contrast, this thesis investigates the effect of switching under active RIS conditions [9][10]. Previously, Xie *et al.* [9] studied and analyzed the switching mechanism of RIS elements and evaluated its impact on EE performance in both passive and active RIS-assisted communication networks. They developed two algorithms based on alternating optimization (AO), employing variable substitution, quadratic transformation, and successive convex approximation (SCA) methods to obtain correspond-

ing suboptimal solutions. Chen *et al.* [10] proposed the BPOF scheme, which demonstrates a beamforming optimization strategy based on communication-element switch control, applied to EE-optimized RIS-assisted wireless systems.

2.2 Motivation

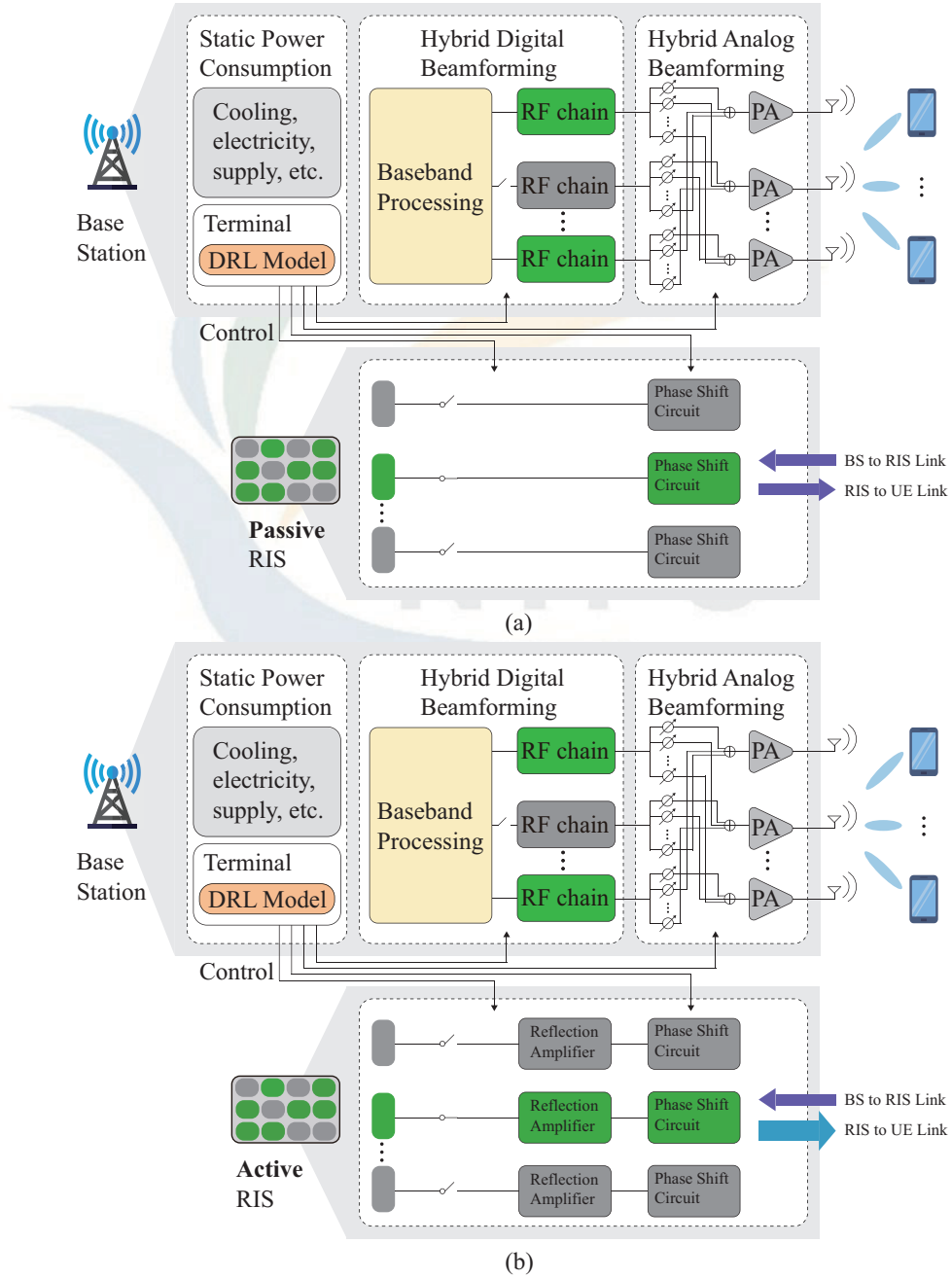


Figure 2.1: (a) Passive RIS architecture diagram and (b) active RIS architecture diagram.

In current wireless communication system research, active reconfigurable intelligent surfaces (active RIS) are regarded as a key technology to overcome the limitations of traditional passive RIS in terms of energy efficiency (EE) and transmission performance, due to their ability to provide active gain. However, the components required for active RIS, such as radio frequency (RF) amplifiers, introduce additional power consumption, which may prevent the overall EE of the system from improving. To address this issue, recent studies have begun exploring the design of active RIS systems incorporating element switching control mechanisms. By selectively activating or deactivating certain RIS elements, unnecessary power consumption can be reduced, thereby enhancing EE. Nevertheless, most existing research relies on mathematical optimization techniques, such as alternating optimization (AO), or successive convex approximation (SCA) [9, 10]. Although these methods are theoretically sound, they face challenges in implementation and computational complexity when applied to dynamic channel environments or high-dimensional control scenarios.

On the other hand, deep reinforcement learning (DRL) has been proven effective in addressing RIS element control, phase adjustment, and beamforming without requiring precise channel models [17]. DRL is particularly suitable for scenarios involving unknown models, high-dimensional state spaces, and dynamic environments. However, current applications of DRL in RIS systems remain limited to passive architectures [19, 20], and research on DRL-based optimization for active RIS combined with element switching control has yet to emerge. Therefore, this study aims to fill this research gap by proposing a DRL-based optimization framework for active RIS, which jointly considers RIS element switching control and transmit beamforming design to effectively improve energy efficiency of the system.

This thesis considers the same two types of communication system architectures as in BPOF [10]. They are the switching control of passive RIS-assisted communication system and the switching control of active RIS-assisted communication system. The former enhances transmission efficiency by adjusting the phase shifts of passive reflecting elements and reduces power consumption by controlling the ON/OFF states of the elements to improve

energy efficiency, as illustrated in Fig. 2 (a). The latter further incorporates active amplifier components to provide additional signal gain, thereby overcoming the multiplicative fading limitations faced by conventional passive RIS, as illustrated in Fig. 2 (b). By comparing the performance of these two architectures under different channel conditions and element switching configurations, this thesis aims to explore the potential benefits of element switching control in active RIS architectures and to design reinforcement learning algorithms suitable for high-dimensional continuous action spaces, with the goal of optimizing energy efficiency.



Chapter 3

Preliminaries

This section 3.1 describes the system model, section 3.2 formulates the problem statement, section 3.3 introduces basic idea, and section 3.4 compares the differences between Catfish-DRL and competing models.

3.1 System Model

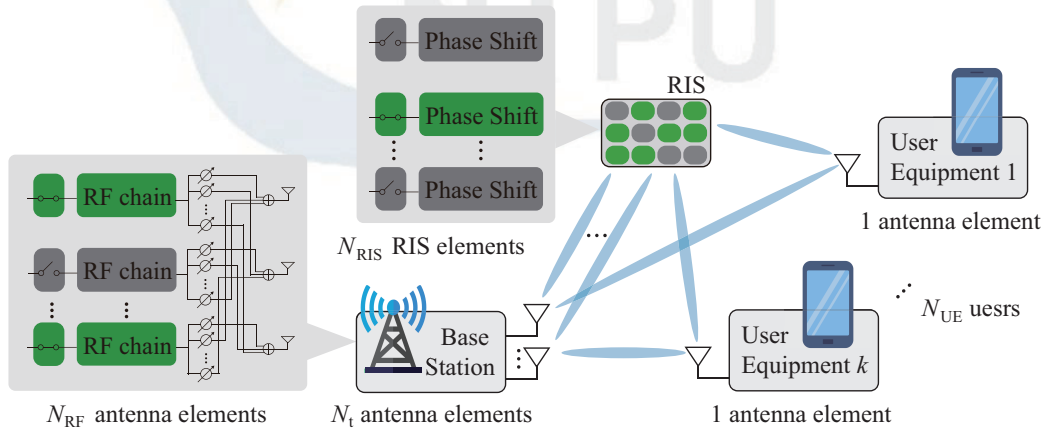


Figure 3.1: A downlink communication system design.

In this work, a hybrid active/passive reconfigurable intelligent surface (RIS) is considered in a multi-user multiple-input single-output (MU-MISO) for downlink communication system. The base station (BS) is assumed to employ a hybrid beamforming (HBF) architectures [10, 13, 14]. Let N_{UE} denote the numbers of single-antenna users, N_{RIS} the numbers of the RIS elements, N_t the numbers of transmit antennas, N_{RF} the numbers of RF chains,

where $N_{\text{RF}} \leq N_t$. As illustrated in Fig. 3, a total of N_{UE} single-antenna users are served by a BS featuring with N_t transmit antennas and N_{RF} switchable radio frequency (RF) chains. Additionally, the communication environment is supported by an active or passive RIS composed of N_{RIS} switchable reflecting elements. The RIS provides an indirect transmission path that complements the direct link, thereby enhancing or extending the system's communication capabilities.

A. Hybrid beamforming model

Consider a hybrid beamforming architecture at the base station that combines a baseband digital precoder with an RF analog beamformer. We adopt the notation as defined in our previous work [10]. The base station in this system consists of three components. Let $F^{\text{RF}} \in \mathbb{C}^{N_t \times N_{\text{RF}}}$ denote the RF analog beamformer, which controls the beam direction. Let $w_k^{\text{BB}} \in \mathbb{C}^{N_{\text{RF}} \times 1}$ denote the baseband digital precoder, which precodes the information transmitted to the k -th user, where $1 \leq k \leq N_{\text{UE}}$. Finally, the switch controller of the RF chains is represented by a binary diagonal matrix $C^{\text{RF}} = \text{diag}(c_1^{\text{RF}}, \dots, c_i^{\text{RF}}, \dots, c_{N_{\text{RF}}}^{\text{RF}}) \in \mathbb{C}^{N_{\text{RF}} \times N_{\text{RF}}}$. The switch $c_i^{\text{RF}} \in \mathbb{R}$ of the i -th RF chain can be defined as:

$$c_j^{\text{RF}} = \begin{cases} 1, & \text{if RF chain } j \text{ in RIS is turned on} \\ 0, & \text{if RF chain } j \text{ in RIS is turned off.} \end{cases} \quad (3.1)$$

where $1 \leq i \leq N_{\text{RF}}$. This matrix determines whether the signal generated by the baseband digital precoder w_k^{BB} for the k -th user is forwarded to the RF analog beamformer F^{RF} for analog weighting (e.g., via phase shifters).

Therefore, the transmit matrix of the base station for the k -th user can be expressed as:

$$x_k = F^{\text{RF}} C^{\text{RF}} w_k^{\text{BB}} s_k \quad (3.2)$$

Here, s_k denotes the data symbol transmitted by the base station to the k -th user. Here, we assume a unit-power symbol, set $\mathbb{E}\{|s_k|^2\} = 1$.

B. Passive RIS and active RIS models

Following the method described in [10], consider three component matrices constituting the reconfigurable intelligent surface (RIS) in the system: Let $\phi^{\text{RIS}} = \text{diag}(e^{j\theta_1}, \dots, e^{j\theta_n}, \dots, e^{j\theta_{N_{\text{RIS}}}}) \in \mathbb{C}^{N_{\text{RIS}} \times N_{\text{RIS}}}$ denote the diagonal phase shift matrix controlling the direction of signal transmission, where the phase shift coefficient $\theta_n \in [0, 2\pi)$, $j = \sqrt{-1}$ and $1 \leq n \leq N_{\text{RIS}}$. $\Lambda^{\text{RIS}} = \text{diag}(\rho_1, \dots, \rho_n, \dots, \rho_{N_{\text{RIS}}}) \in \mathbb{C}^{N_{\text{RIS}} \times N_{\text{RIS}}}$ denote the power amplification (PA) diagonal matrix specific to active RIS, where $1 \leq n \leq N_{\text{RIS}}$ and $\rho_n \geq 0$. Since each element has an amplification factor, it can consume additional energy to boost the signal, similarly to a relay base station. Finally, the switch controller of the intelligent reflecting elements in passive and active modes are modeled respectively as binary diagonal matrices $B^{\text{RIS}} = \text{diag}(\beta_1^{\text{RIS}}, \dots, \beta_n^{\text{RIS}}, \dots, \beta_{N_{\text{RIS}}}^{\text{RIS}}) \in \mathbb{C}^{N_{\text{RIS}} \times N_{\text{RIS}}}$ and $A^{\text{RIS}} = \text{diag}(\alpha_1^{\text{RIS}}, \dots, \alpha_n^{\text{RIS}}, \dots, \alpha_{N_{\text{RIS}}}^{\text{RIS}}) \in \mathbb{C}^{N_{\text{RIS}} \times N_{\text{RIS}}}$. The switch $\beta_n^{\text{RIS}} \in \mathbb{R}$ of the n -th passive RIS element and the switch $\alpha_n^{\text{RIS}} \in \mathbb{R}$ of the n -th active RIS element can be defined as follows, respectively:

$$\begin{cases} \beta_n^{\text{RIS}} = \begin{cases} 1, & \text{if RIS element } n \text{ in BS is turned on} \\ 0, & \text{if RIS element } n \text{ in BS is turned off.} \end{cases} \\ \alpha_n^{\text{RIS}} = \begin{cases} 1, & \text{if RIS element } n \text{ in BS is turned on} \\ 0, & \text{if RIS element } n \text{ in BS is turned off.} \end{cases} \end{cases} \quad (3.3)$$

Indicating that then i -th RIS element is activated when set to 1 and deactivated otherwise, with $1 \leq n \leq N_{\text{RIS}}$. This matrix determines whether each RIS element performs effective reflection and can be used to adjust power consumption in both active and passive RIS scenarios.

C. Signal transmission model for passive and active RIS

Assuming that all involved channels are perfectly known at the BS [8], the system considers three communication channel matrices for each $k \in [1, 2, \dots, N_{\text{UE}}]$. The direct channel between the base station and the k -th user is denoted as $g_k \in \mathbb{C}^{1 \times N_t}$. The channel between the base station and the reconfigurable intelligent surface is denoted as $Q \in \mathbb{C}^{N_{\text{RIS}} \times N_t}$. The

channel between the RIS and the k -th user is denoted as $h_k \in \mathbb{C}^{1 \times N_{\text{RIS}}}$. Let x_k denote the transmit signal vector of the base station. The base station transmits signals to the users through both the direct channel g_k and the indirect channel via the RIS. This thesis models the channel matrices Q, h_k and g_k corresponding to the BS–RIS, RIS–UE, and BS–UE links, using Rician fading [8]:

$$\begin{cases} Q = \sqrt{\frac{L^{\text{BR}} K^{\text{BR}}}{K^{\text{BR}} + 1}} Q^{\text{LoS}} + \sqrt{\frac{L^{\text{BR}}}{K^{\text{BR}} + 1}} Q^{\text{NLoS}} \\ h_k = \sqrt{\frac{L_k^{\text{RU}} K^{\text{RU}}}{K^{\text{RU}} + 1}} h_k^{\text{LoS}} + \sqrt{\frac{L_k^{\text{RU}}}{K^{\text{RU}} + 1}} h_k^{\text{NLoS}}, \quad k \in N_{\text{UE}} \\ g_k = \sqrt{\frac{L_k^{\text{BU}} K^{\text{BU}}}{K^{\text{BU}} + 1}} g_k^{\text{LoS}} + \sqrt{\frac{L_k^{\text{BU}}}{K^{\text{BU}} + 1}} g_k^{\text{NLoS}} \end{cases} \quad (3.4)$$

Here, LoS and NLoS represent the line-of-sight and non-line-of-sight components, respectively. For the NLoS components of Q, h_k and g_k , i.e. $Q^{\text{NLoS}} \sim \mathbb{CN}(0, 1), h_k^{\text{NLoS}} \sim \mathbb{CN}(0, 1), g_k^{\text{NLoS}} \sim \mathbb{CN}(0, 1), k \in N_{\text{UE}}$, they are assumed to be independently and identically distributed (i.i.d.) complex circularly symmetric Gaussian random variables with zero mean and unit variance. L^{BR} denotes the large-scale path loss coefficient for the BS–RIS link; L_k^{RU} and L_k^{BU} are the large-scale path loss coefficients for the RIS–UE and BS–UE links for the k -th UE, respectively. $K^{\text{BR}}, K^{\text{RU}}$ and K^{BU} denote the corresponding Rician K -factors.

Therefore, the received transmission signal at the k -th user can be expressed, under the passive and active RIS modes respectively, as follows:

$$\begin{cases} y_k^{\text{pas}} = \sum_{k=1}^{N_{\text{UE}}} (g_k + h_k \phi^{\text{RIS}} B^{\text{RIS}} Q) x_k + n_k \\ y_k^{\text{act}} = \sum_{k=1}^{N_{\text{UE}}} (g_k + h_k \phi^{\text{RIS}} \Lambda^{\text{RIS}} A^{\text{RIS}} Q) x_k + h_k \phi^{\text{RIS}} \Lambda^{\text{RIS}} A^{\text{RIS}} z + n_k \end{cases} \quad (3.5)$$

Here, $n_k \sim \mathbb{CN}(0, \delta^2)$ denotes the additive white Gaussian noise (AWGN) at the receiver of the k -th user, with zero mean and variance δ^2 . $z \sim \mathbb{CN}(0, \sigma^2)$ represents the thermal noise introduced by the active RIS, also modeled as AWGN with zero mean and variance σ^2 , which arises due to the signal amplification process performed by the active reconfigurable intelligent

surface.

The signal-to-interference-plus-noise ratio (SINR) for the k -th user can be expressed under the passive and active modes, respectively, as follows:

$$\begin{cases} \zeta_k^{pas} = \frac{|(g_k + h_k \phi^{RIS} B^{RIS} Q)_{x_k}|^2}{\sum_{k'=1, k' \neq k}^{N_{UE}} |(g_k + h_k \phi^{RIS} B^{RIS} Q)_{x_{k'}}|^2 + \delta^2} \\ \zeta_k^{act} = \frac{|(g_k + h_k \phi^{RIS} \Lambda^{RIS} A^{RIS} Q)_{x_k}|^2}{\sum_{k'=1, k' \neq k}^{N_{UE}} |(g_k + h_k \phi^{RIS} \Lambda^{RIS} A^{RIS} Q)_{x_{k'}}|^2 + \sigma^2 \|h_k \phi^{RIS} \Lambda^{RIS} A^{RIS}\|_F^2 + \delta^2} \end{cases} \quad (3.6)$$

Here, $\|\cdot\|_F$ denotes the squared Frobenius norm (L2 norm), representing the total energy of all elements in a matrix. The entire expression is defined as the ratio between the power of the desired signal and the total power of interference from other users. The term δ^2 represents the AWGN power at the receiver. Under the active RIS mode, the active elements introduce additional amplified thermal noise, represented by $\sigma^2 \|h_k \Phi^{RIS} \Lambda^{RIS} A^{RIS}\|_F^2$, which measures the total energy transferred through the active reconfigurable intelligent surface.

According to Shannon's theorem, the spectral efficiency (SE) of the base station for the k -th user under passive and active modes is expressed as follows, respectively:

$$\begin{cases} R_k^{pas}(\mathbf{w}_k^{BB}, \mathbf{F}^{RF}, \mathbf{C}^{RF}, \phi^{RIS}, \mathbf{B}^{RIS}) = BW \cdot \log_2(1 + \zeta_k^{pas}), \\ R_k^{act}(\mathbf{w}_k^{BB}, \mathbf{F}^{RF}, \mathbf{C}^{RF}, \Lambda^{RIS}, \phi^{RIS}, \mathbf{A}^{RIS}) = BW \cdot \log_2(1 + \zeta_k^{act}). \end{cases} \quad (3.7)$$

Here, the bandwidth BW is set to 1 Hz.

C. Power model

Considering this thesis's RIS-assisted communication system, its energy consumption can be divided into non-transmission power consumption and transmission power consumption. P^{static} is defined as the static power consumed by the electronic components of each device, while P^{dynamic} refers to the transmission power.

Therefore, the total power consumption of the system is expressed as:

$$P^{\text{total}} = P^{\text{static}} + P^{\text{dynamic}} \quad (3.8)$$

Here, the transmission power consumption can also be divided into two parts, with the base station's contribution expressed as:

The base station's power consumption is given by

$$P^{BS} = \|C^{RF}\|_0 P^{RF} + \sum_{k=1}^{N_{UE}} \|F^{RF} C^{RF} w_k^{BB}\|_F^2 \quad (3.9)$$

where P^{RF} indicates the fixed power consumption per RF chain. $\|\cdot\|_0$ represents the L0 norm, which is defined as the number of non-zero elements in a vector.

The power consumption of the reconfigurable intelligent surface is expressed under the passive and active modes, respectively, as follows:

$$\begin{cases} P^{\text{pas}} = \|C^{RIS}\|_0 P_C, \\ P^{\text{act}} = \|C^{RIS}\|_0 (P_C + P_{DC}) + \sum_{k=1}^{N_{UE}} \left\| \phi^{RIS} \Lambda^{RIS} A^{RIS} Q^{RF} F^{RF} C^{RF} w_k^{BB} \right\|_F^2 + \sigma^2 \left\| \phi^{RIS} \Lambda^{RIS} A^{RIS} \right\|_F^2 \end{cases} \quad (3.10)$$

Here, P_C denotes the power consumption of a single RIS element, and P_{DC} represents the DC bias power consumption.

After obtaining the fixed and transmission power consumption of all communication devices, the system's total energy consumption is expressed under the passive and active modes, respectively, as follows:

$$\begin{cases} P_{\text{Tot}}^{\text{pas}}(w_k^{BB}, F^{RF}, C^{RF}, \phi^{RIS}, B^{RIS}) = P^{\text{static}} + P^{BS} + P^{\text{pas}}, \\ P_{\text{Tot}}^{\text{act}}(w_k^{BB}, F^{RF}, C^{RF}, \Lambda^{RIS}, \phi^{RIS}, A^{RIS}) = P^{\text{static}} + P^{BS} + P^{\text{act}}. \end{cases} \quad (3.11)$$

3.2 Problem formulation

The objective of this section is to formulate a transmission energy efficiency (EE) problem suitable for the Catfish-effect-based deep reinforcement learning (CDRL) framework. The goal is to maintain a certain transmission rate while precisely adjusting the BS beamforming and RIS phase shifts, minimizing the number of active elements, and maximizing the system's

energy efficiency to achieve the ultimate objective of energy-efficient communications.

Considering the environment where the base station serves all users with the assistance of a passive reconfigurable intelligent surface, the system energy efficiency optimization problem can be defined as [10]:

$$P_0 : \max_{w_k^{BB}, F^{RF}, C^{RF}, \phi^{RIS}, B^{RIS}} \frac{\sum_{k=1}^{N_{UE}} R_k^{pas}(w_k^{BB}, F^{RF}, C^{RF}, \phi^{RIS}, B^{RIS})}{P_{Tot}^{pas}(w_k^{BB}, F^{RF}, C^{RF}, \phi^{RIS}, B^{RIS})} \quad (3.12a)$$

$$\text{s.t. } C_1 : R_k^{pas} \geq R^{min}, \quad (3.12b)$$

$$C_2 : \sum_{k=1}^{N_{UE}} \|F^{RF} C^{RF} w_k^{BB}\|^2 \leq P_{BS}^{max}, \quad (3.12c)$$

$$C_3 : |F_{i,j}^{RF}| = 1, \quad (3.12d)$$

$$C_4 : |\theta_n| = 1, \quad (3.12e)$$

$$C_5 : c_j^{RF} \in \{0, 1\}, \quad (3.12f)$$

$$C_6 : \beta_n^{RIS} \in \{0, 1\}, \quad (3.12g)$$

$$\forall j \in N_{RF}, n \in N_{RIS}, k \in N_{UE}, i \in N_t.$$

Considering the environment where the base station serves all users with the assistance of an active reconfigurable intelligent surface, the system energy efficiency optimization problem can be defined as:

$$P_1 : \max_{\{w_k^{BB}, F^{RF}, C^{RF}, \Lambda^{RIS}, \phi^{RIS}, A^{RIS}\}} \frac{\sum_{k=1}^{N_{UE}} R_k^{act}(w_k^{BB}, F^{RF}, C^{RF}, \Lambda^{RIS}, \phi^{RIS}, A^{RIS})}{P_{Tot}^{act}(w_k^{BB}, F^{RF}, C^{RF}, \Lambda^{RIS}, \phi^{RIS}, A^{RIS})} \quad (3.13a)$$

$$\text{s.t. } C_1 : R_k^{act} \geq R^{min}, \quad (3.13b)$$

$$C_2 : \sum_{k=1}^{N_{UE}} \|F^{RF} C^{RF} \mathbf{w}_k^{BB}\|^2 \leq P_{BS}^{max}, \quad (3.13c)$$

$$C_3 : |F_{i,j}^{RF}| = 1, \quad (3.13d)$$

$$C_4 : |\theta_n| = 1, \quad (3.13e)$$

$$C_5 : c_j^{RF} \in \{0, 1\}, \quad (3.13f)$$

$$C_6 : \alpha_n^{RIS} \in \{0, 1\}, \quad (3.13g)$$

$$C_7 : \rho_n \geq 0, \quad (3.13h)$$

$$C_8 : \sum_{k=1}^{N_{UE}} \|\phi^{RIS} \Lambda^{RIS} A^{RIS} Q^{RF} F^{RF} C^{RF} w_k^{BB}\|^2 + \sigma^2 \|\phi^{RIS} \Lambda^{RIS} A^{RIS}\|_F^2 \leq P_{RIS}^{max} \quad (3.13i)$$

$$\forall j \in N_{RF}, n \in N_{RIS}, k \in N_{UE}, i \in N_t.$$

C_1 : When the base station serves the k -th user, the transmission rate R_k^{pas} / R_k^{act} must be maintained above the predefined threshold R^{min} to ensure user experience. C_2 : The total transmission power when the base station serves the k -th user, expressed as $\sum_{k=1}^{N_{UE}} \|F^{RF} C^{RF} w_k^{BB}\|_F^2$, must remain below the power budget P_{BS}^{max} . C_3 : Each element in F^{RF} is constrained to have unit magnitude, as F^{RF} is responsible only for beam direction control. C_4 : The phase shift θ_n of the n -th reflective element is constrained within $[0, 2\pi)$, modeling the periodic phase control capability of the phase shifters. C_5 : The RF-chain switch variable c_j^{RF} at the base station must remain in $\{0, 1\}$, where 1 indicates the link is active, and 0 indicates it is inactive. C_6 : The RIS switch variables β_n^{RIS} and α_n^{RIS} must remain in $\{0, 1\}$, where 1 indicates the reflecting element is active, and 0 indicates it is inactive. C_7 : The amplifier gain ρ_n at the active RIS must be constrained to non-negative real values, as each power amplifier (PA) can only provide non-negative gain. C_8 : When the RIS serves the k -th user, the total transmission power, expressed as $\sum_{k=1}^{N_{UE}} \|\phi^{RIS} \Lambda^{RIS} A^{RIS} Q^{RF} F^{RF} C^{RF} w_k^{BB}\|_F^2 + \sigma^2 \|\phi^{RIS} \Lambda^{RIS} A^{RIS}\|_F^2$, must

remain below the predefined threshold $P_{RIS}^{\max}$.

3.3 Basic idea

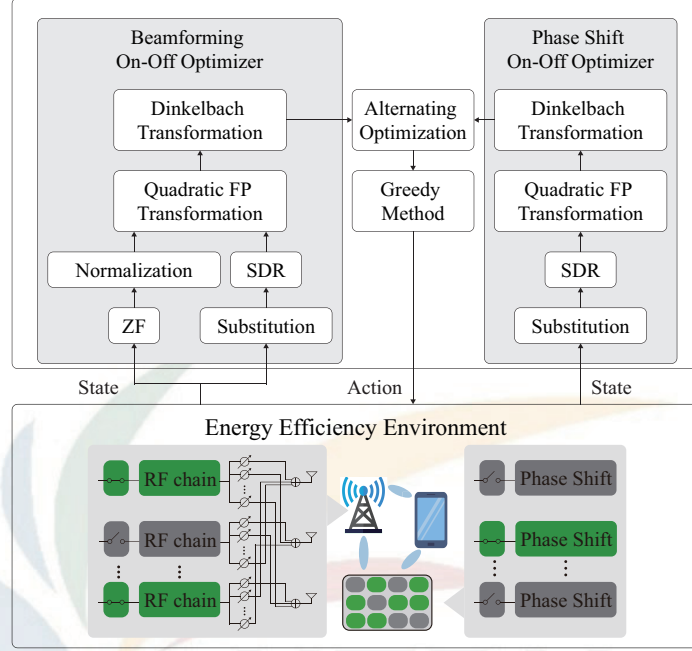


Figure 3.2: Basic idea of BPOF.

In [10], Chen et al. formulated a maximization problem based on passive and active RIS systems. To mathematically address the multiple coupled optimization variables, the Dinkelbach method and quadratic transformation were adopted, and alternating optimization (AO) was applied to approximate the optimal solution, as illustrated in Fig. 4. In [11], Huang et al. proposed a DRL-based algorithm, where the state s_t is input to the reinforcement learning actor network to generate the resulting action a_t . The environment then transitions to the next state and returns a reward r_t based on a_t , while the critic network updates the actor policy using the tuple (s_t, a_t, r_t, s_{t+1}) , as shown in Fig. 5. In [21], Zheng et al. introduced a teacher-student learning framework that leverages the teacher's domain knowledge to improve the robustness of the student's learning. They developed a confidence-check mechanism for identifying critical states, reward shaping for injecting teaching signals into the student network, and prioritized experience replay to address the challenge of limited teacher data, as

illustrated in Fig. 6.

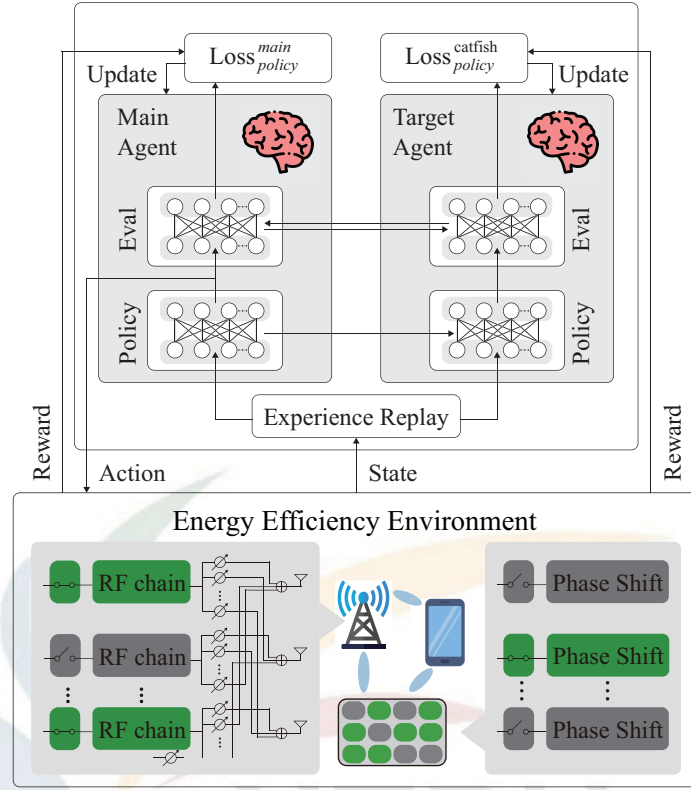


Figure 3.3: Basic idea of Actor-Critic.

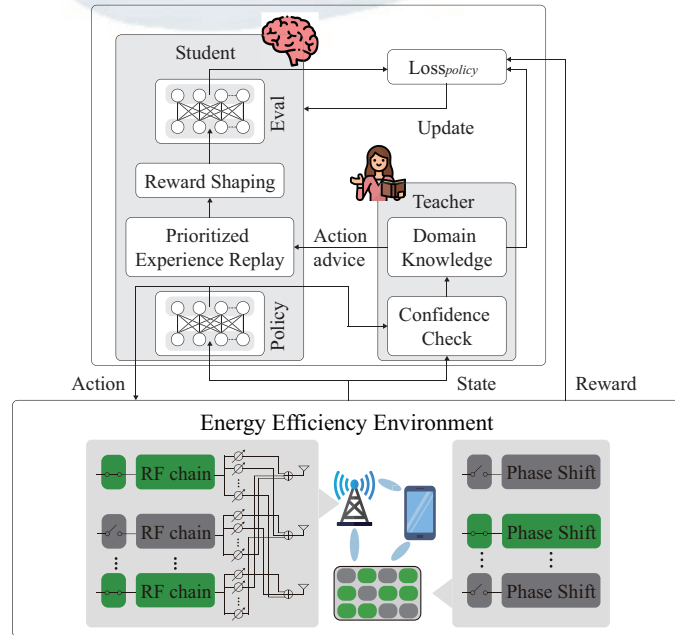


Figure 3.4: Basic idea of Teacher-Student Learning.

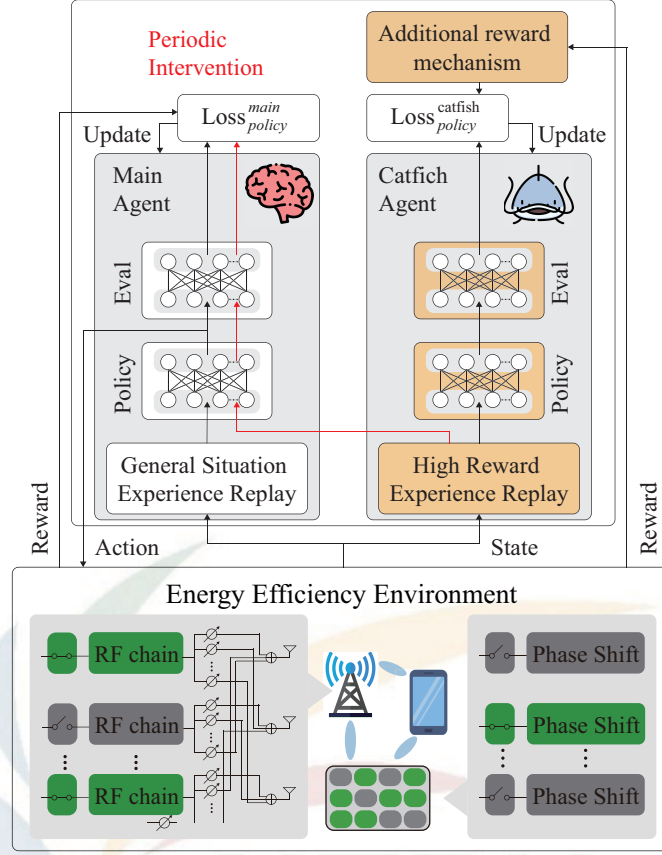


Figure 3.5: Basic idea of CDRL.

After formulating the energy efficiency (EE) maximization problem P_0 and P_1 , this section proposes a deep reinforcement learning-based framework to solve it. As illustrated in Fig. 7, this study introduces an innovative Catfish-effect-based deep reinforcement learning (CDRL) architecture, which enhances sample efficiency through a dual-agent, dual-replay mechanism. Inspired by the Catfish effect in management—where a newly introduced competitor stimulates existing members to perform better—the framework deploys a catfish agent dedicated to exploring and collecting rare, high-reward experiences in the EE-optimal region, which the high-reward experience replay buffer holds. In parallel, the main agent interacts with the environment under general conditions and accumulates data in the general situation experience replay buffer. At random intervals, the system triggers intervention by injecting high-reward batches filtered by the catfish agent into the main agent’s training process. This approach enables experience-level knowledge transfer, which differs from traditional Teacher–Student

Learning based on weight or logit distillation (as shown in Fig. 6). It avoids strategy stagnation due to over-imitation and preserves policy diversity between the two agents.

3.4 Comparison of Catfish-DRL with adversarial models

Table 3.1: Comparison of Catfish-DRL with adversarial models.

Learning models	Advantages	Limitations
Actor-Critic [11]	Suitable for continuous action spaces. Does not rely on labeled samples. Low implementation complexity and highly scalable.	Actor and critic are interdependent, leading to training instability.
BPOF [10]	Alternating iterative decomposition plus a greedy post-processing step converts relaxed weights into concrete on/off decisions, reducing combinatorial complexity.	Still a heuristic approximation of a non-convex problem, incurring high measurement and synchronization overhead in practice.
Teacher-Student [21]	Instructor guidance can improve empirical performance and stability, while preserving students' autonomous exploration.	Relies on heuristic domain rules whose inaccuracies can misguide learning, and the teacher's simplified design may cap the ultimate performance ceiling.
Catfish DRL	Dual agents with partitioned replay buffers sharpen exploration-exploitation and accelerate convergence toward high-quality (high EE) regions.	Increases implementation, computational, and memory costs, and exhibits greater instability during training.

Table 3.1 shows the advantages and limitations of several competitor models relative to CDRL, including Actor-Critic [11], BPOF [10] and Teacher-Student [21]. The Actor-Critic [11] architecture combines policy-gradient and value-based methods, supports single-step updates and is suitable for continuous action spaces, learns through interaction with the environment and therefore does not rely on labeled samples, and has low implementation complexity that is easily extended to various reinforcement-learning improvement. However, the actor and critic alternate updates and depend on each other, ultimately leading to unstable training.

BPOF [10] employs an alternating iterative optimization framework that jointly adjusts ZF, SDR and switch-weight parameters, uses a greedy method in the final stage to convert relaxed weights into actual on/off decisions to reduce search cost, but remains a heuristic approximation to a nonconvex problem and incurs high measurement and synchronization overhead in practice. In Teacher-Student learning [21], teacher provides advice to student at critical states to reduce performance disparity and achieve tail-end performance improvement. Instructor guidance can improve empirical performance and stability while preserving the student’s autonomous exploration ability, yet relies on heuristic domain rules whose inaccuracies may mislead learning and whose simplified design may limit the ultimate performance ceiling. This thesis’s CDRL adopts a strategy in which the catfish agent randomly perturbs the main agent during training to encourage the main agent to explore unvisited action spaces and avoid local optima. In other words, CDRL’s strategy may cause larger fluctuations in the reward curve during training, but under its high-risk, high-reward characteristic it has greater chance to achieve better energy-efficiency performance. Similar to BPOF [10], it presents higher implementation, computational and memory-cost issues.

Chapter 4

Catfish-effect-based DRL for Energy-Efficient Joint Switching

In this chapter, we propose a catfish-based deep reinforcement learning algorithm to optimize the energy efficiency of communication networks. The flowchart of the proposed CDRL algorithm is shown in Fig. 8, and the content of each phase is as follows:

1. ***Catfish-effect-based data preparation phase***: The first stage is data preprocessing, aimed at preparing the training data for the subsequent deep reinforcement learning phase by generating communication states in advance for experience replay with different EE criteria.
2. ***Catfish-effect-based DRL training phase***: The second stage is the general training of the catfish deep reinforcement learning with competitive periodic interventions, designed to transfer high-reward knowledge from the catfish agent to the main agent and to stimulate the main agent to explore and pursue better optimization strategies.
3. ***Catfish-effect-Based DRL optimization of energy efficiency phase***: The third stage uses the trained CDRL policy to optimize communication EE.

4.1 Catfish-effect-based data preparation

In this phase, the pre-generated simulated communication data will be referenced by the designed CDRL agents, enabling the CDRL to learn exemplary solution cases and derive

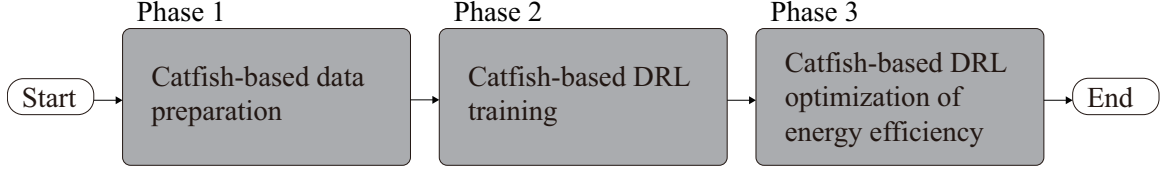


Figure 4.1: The flowchart of the proposed energy efficiency optimization.

optimal solutions for subsequent problems. The communication environment states to be learned include the digital beamforming matrix w_k^{BB} , the RF beamforming matrix F^{RF} , the RF-chain switch matrix C^{RF} , the active RIS power amplification matrix Λ^{RIS} , the phase shift matrix ϕ^{RIS} , the passive RIS element switch matrix B^{RIS} , and the active RIS element switch matrix A^{RIS} . These parameters represent the smallest adjustable units in the entire communication network. These six states are input to the evaluation network to enhance the communication quality output by the main agent, as shown in Fig. 4.2.

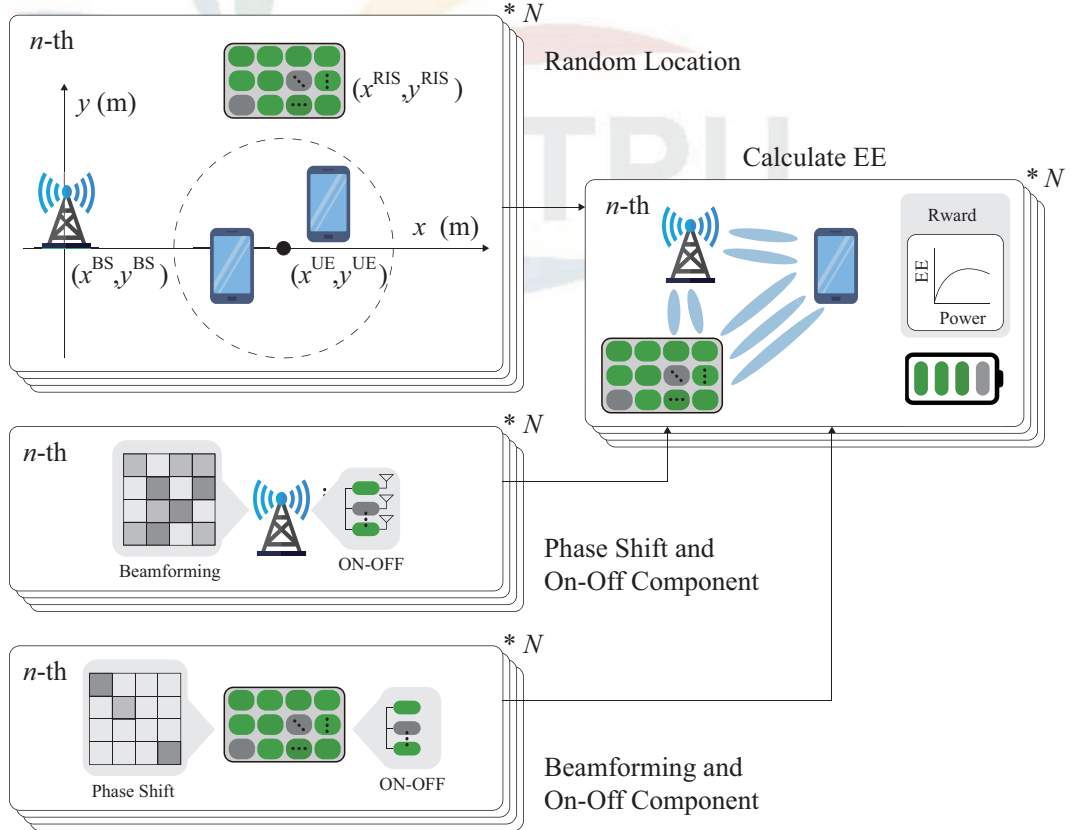


Figure 4.2: The flowchart of prepare training data.

As illustrated in Fig. 9, efficient communication data is pre-generated to build the experi-

ence memory for CDRL. We first randomly generate the spatial layout of the BS, UEs, and RIS to obtain the three channel matrices Q , h_k , and g_k , referring to formula (4). For RIS-related matrix generation, we first process the passive RIS. Considering N -bit phase shifters, a set of candidate phase shifts $\tilde{\phi}_m^{RIS}$, where $1 \leq m \leq 2^N$, is generated using the discrete Fourier transform (DFT) codebook $\tilde{\Phi}^{RIS} = \{\tilde{\phi}_1^{RIS}, \dots, \tilde{\phi}_m^{RIS}, \dots, \tilde{\phi}_{2^N}^{RIS}\}$ [13]. For each candidate phase shift $\tilde{\phi}_m^{RIS}$, it is multiplied by the corresponding random RF-chain switch matrix $\tilde{B}_m^{RIS}$, resulting in candidate passive RIS phase shifts $\tilde{\phi}_m^{RIS} \tilde{B}_m^{RIS}$ with corresponding RIS elements turned off. For active RIS, a uniformly random active RIS power amplification matrix set $\hat{\Lambda}^{RIS} = \{\hat{\Lambda}_1^{RIS}, \dots, \hat{\Lambda}_m^{RIS}, \dots, \hat{\Lambda}_{2^N}^{RIS}\}$, where $1 \leq m \leq 2^N$ and $\hat{\Lambda}_m^{RIS} \geq 1$, is generated. Finally, the candidate active RIS phase shifts with corresponding RIS elements turned off are expressed as $\tilde{\phi}_m^{RIS} \hat{\Lambda}_m^{RIS} \tilde{A}_m^{RIS}$.

For the BS part, similarly, a set of candidate beams $\tilde{F}_m^{RF}$, where $1 \leq m \leq 2^N$, is generated using the discrete Fourier transform (DFT) codebook $\tilde{F}^{RF} = \{\tilde{F}_1^{RF}, \dots, \tilde{F}_m^{RF}, \dots, \tilde{F}_{2^N}^{RF}\}$. For each candidate beam $\tilde{F}_m^{RF}$, it is multiplied by the corresponding random RF-chain switch matrix $\tilde{C}_m^{RF}$, yielding candidate beams $\tilde{F}_m^{RF} \tilde{C}_m^{RF}$ with corresponding RF-chain elements turned off. Then, the weighted minimum mean square error (WMMSE) algorithm [22] is used to compute the corresponding digital baseband precoding matrix $\tilde{w}_{k,m}^{BB}$. Next, the spectral efficiency (SE) formula (6) calculates the user rates R_k^{pas} and R_k^{act} . Combining the power amplifier consumption and the energy consumption of activated RF chains, the overall energy efficiency (EE) is calculated, referring to formulas (12) and (13). Finally, the candidate with the maximum EE is selected as the final analog precoding matrix $\hat{F}^{RF}$, as illustrated in Fig. 10.

After completing all steps, the system state, selected beam index, and EE triplet are stored in the replay buffer, providing accurate and consistent training samples for subsequent deep reinforcement learning. In phase 4.2.2, the generated data can be stored into the catfish replay memory according to matching thresholds and is represented as $D = \{D_1, \dots, D_n, \dots, D_N\}$, where $1 \leq n \leq N$, serving as the initial experience replay data for CDRL.

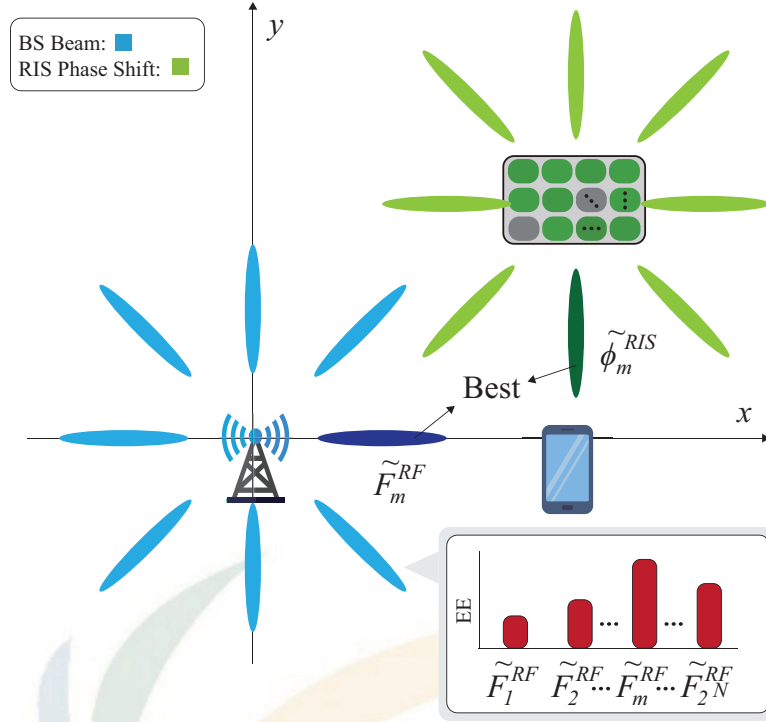


Figure 4.3: Beam generation diagram for prepare training data.

4.2 Catfish-effect-based DRL training

Since the energy efficiency problem discussed in chapter 3 is a non-convex optimization problem, traditional mathematical methods require non-convex-to-convex transformations, resulting in computational complexity [11]. Compared to conventional alternating optimization (AO) methods, deep reinforcement learning (DRL) can optimize complex dynamic problems without requiring prior mathematical relaxation [13]. This phase provides an overview of the proposed CDRL model architecture and training methodology. The main agent is introduced in phase 4.2.1, including the formulation of states $s_t^{\text{pas}}/s_t^{\text{act}}$, actions $a_t^{\text{pas}}/a_t^{\text{act}}$, and rewards $r_t^{\text{pas}}/r_t^{\text{act}}$. The catfish agent is also introduced in phase 4.2.1, covering experience stratification, Asymmetric Discounting Strategy, and the adaptive competitive reward mechanism.

4.2.1 Main agent of Catfish-effect-based DRL

The core of reinforcement learning lies in the interaction between the agent and the environment, where the agent iteratively adjusts its policy based on received rewards, aiming to maximize the cumulative reward. To solve for the optimal EE parameters using DRL, this phase defines the state, action, and reward components [11].

State: used to represent the current status of each component in the environment. The state at time step t , includes the system's total power consumption $P_{\text{Tot}}^{\text{pas}}/P_{\text{Tot}}^{\text{act}}$, the three channel matrices Q , h_k and g_k , and the actions at the time step t , $a_t^{\text{pas}}/a_t^{\text{act}}$. The state at time step t is represented as:

$$\begin{cases} s_t^{\text{pas}} = [P_{\text{Tot}}^{\text{pas}}, Q, h_k, g_k, a_t^{\text{pas}}], \\ s_t^{\text{act}} = [P_{\text{Tot}}^{\text{act}}, Q, h_k, g_k, a_t^{\text{act}}]. \end{cases} \quad (4.1)$$

Action: Represents the operations executed by each component in the current environment. It includes all component state matrices $w_k^{BB}, F^{RF}, C^{RF}, \Lambda^{RIS}, \phi^{RIS}, B^{RIS}/A^{RIS}$. The action at time step t is denoted as:

$$\begin{cases} a_t^{\text{pas}} = [w_k^{BB}, F^{RF}, C^{RF}, \phi^{RIS}, B^{RIS}], \\ a_t^{\text{act}} = [w_k^{BB}, F^{RF}, C^{RF}, \Lambda^{RIS}, \phi^{RIS}, A^{RIS}]. \end{cases} \quad (4.2)$$

Reward: Represents the feedback from the environment in response to a_t . We are using energy efficiency as a reward. The reward at time step t is denoted as:

$$\begin{cases} r_t^{\text{pas}} = \sum_{k=1}^{N_{\text{UE}}} R_k^{\text{pas}}(w_k^{BB}, F^{RF}, C^{RF}, \phi^{RIS}, B^{RIS}), \\ r_t^{\text{act}} = \sum_{k=1}^{N_{\text{UE}}} R_k^{\text{act}}(w_k^{BB}, F^{RF}, C^{RF}, \Lambda^{RIS}, \phi^{RIS}, A^{RIS}). \end{cases} \quad (4.3)$$

- **Critic Network:** The critic network takes the current state and action as input and outputs Q-value, denoted as $Q(s_t, a_t)$, where Φ represents the parameters of the value network.
- **Actor Network:** The actor network receives the current state of the environment as input

and, based on a policy function, determines the action to take. The policy is represented as $\pi_\theta(s_t)$, where θ denotes the parameters of the policy network, which guides the agent's action selection, as illustrated in Fig. 11.

Since the channel matrices and component state matrices are complex-valued while neural networks can only process real numbers, both inputs and outputs are split into real and imaginary parts. To enhance training stability, batch normalization is introduced in the hidden layers, normalizing the outputs of the previous layer to zero mean and unit variance.

The critic evaluates the value of the actions taken by the actor and provides correction signals. This is typically measured using temporal difference (TD) error δ , defined as:

$$\delta_t = r_t + \gamma Q_\Phi(s_{t+1}, \pi_\theta(s_{t+1})) - Q_\Phi(s_t, a_t) \quad (4.4)$$

Here, $\pi_\theta(s_{t+1})$ denotes the action given by the actor at the next state. γ is the discount factor that determines the importance of the future Q-value $Q_\Phi(s_{t+1}, \pi_\theta(s_{t+1}))$, typically ranging between 0.9 and 0.99. Equation 4.4 measures the discrepancy between the current Q-value estimate and the one-step actual reward plus the estimated Q-value at the next step (target Q-value). This quantifies the difference between the estimate and reality, which is used to guide the update of the weighting parameters.

The critic uses a neural network to approximate the Q-value function $Q(s_t, a_t)$, and updates the network parameters by minimizing the mean squared error (MSE) loss, defined as:

$$L(\Phi) = (\delta_t)^2 \quad (4.5)$$

Then, gradient descent is used to update the critic's parameters Φ as follows:

$$\Phi_{t+1} = \Phi_t - \mu_c \nabla_\Phi L(\Phi_t) \quad (4.6)$$

Here, μ_c denotes the learning rate, and ∇ represents the gradient with respect to Φ of the loss function $L(\Phi_t)$.

$$\theta_{t+1} = \theta_t - \mu_a \left[\nabla_a Q_\Phi(s_t, a) \Big|_{a=\pi_\theta(s_t)} \right] \nabla_\theta \pi_\theta(s_t) \quad (4.7)$$

Here, μ_a denotes the learning rate. First, the critic computes the gradient with respect to the action to determine the direction that increases the Q-value. Then, the actor propagates this information back to its parameters θ through its own gradient.

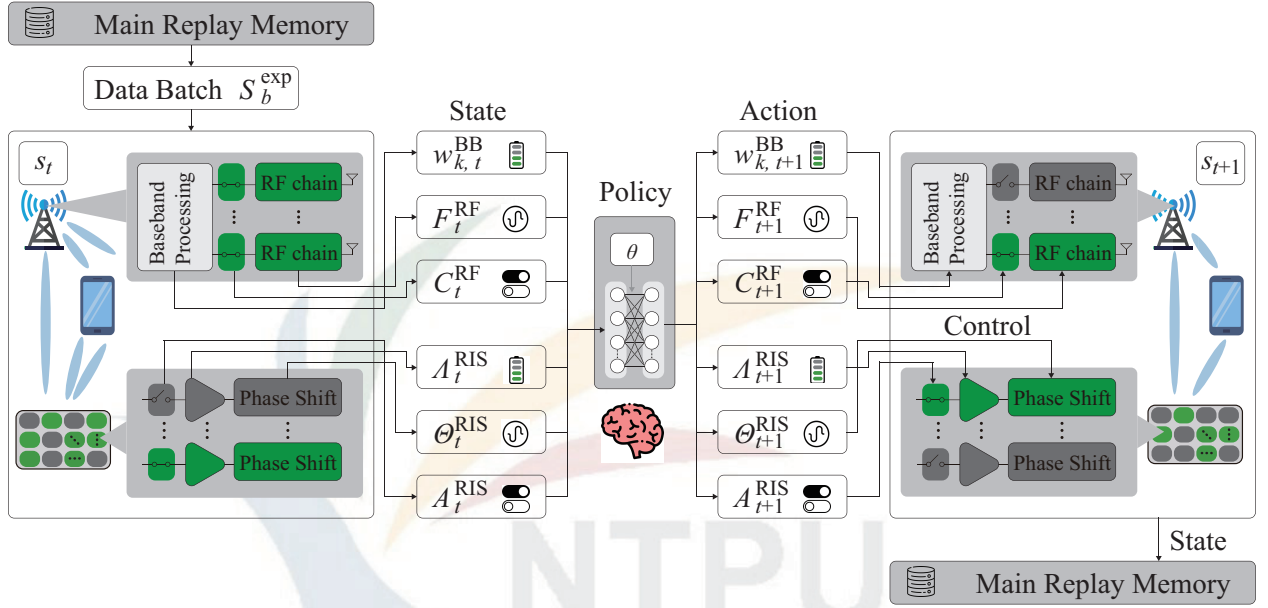


Figure 4.4: The flowchart of main agent (Active RIS).

4.2.2 Catfish agent of Catfish-effect-based DRL

This phase continues the work from phase 4.2.1 by explaining the dedicated mechanisms of the catfish agent and how it collaborates with the main agent. The goal is to achieve both the stimulation of exploration by the main agent and experience-level knowledge transfer. The core design of the catfish agent includes three key aspects, all aimed at ensuring its policy outperforms that of the main agent: first, during the data pre-generation phase in phase 4.1, all experiences are routed based on an energy efficiency (EE) threshold. Second, the discount factor of the catfish agent is set higher to emphasize long-term return estimation. Third, an adaptive competitive reward mechanism is established to encourage the catfish agent to maintain its leading advantage.

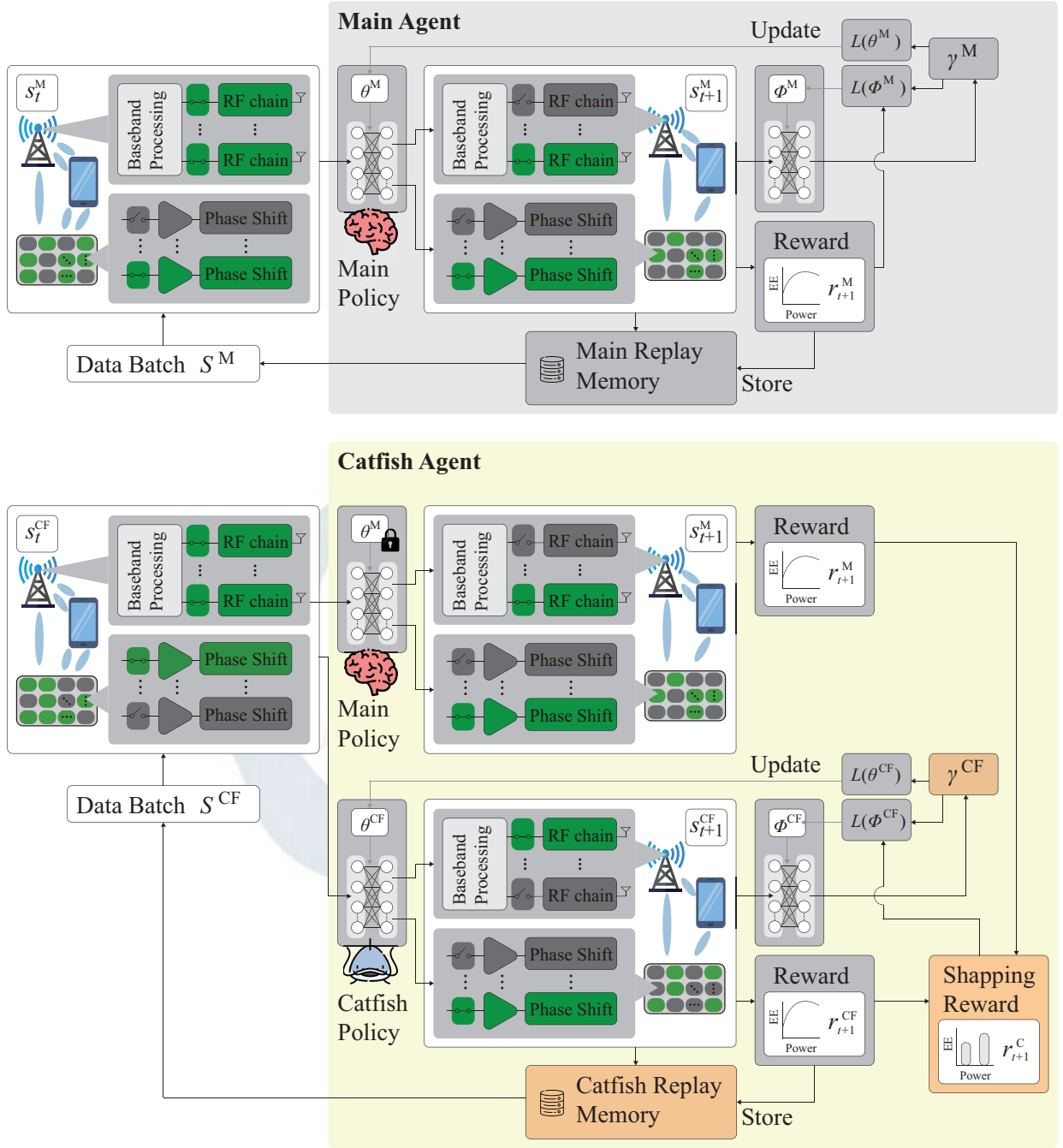


Figure 4.5: General training phase of CDRL.

Next, the two training modes of CDRL are presented. As shown in Fig. 4.5, two agents are established simultaneously: the main agent and the catfish agent. The main agent learns experiences under general energy efficiency (EE) conditions using a smaller discount factor. The catfish agent uses a larger discount factor to learn high EE experiences, aiming to capture

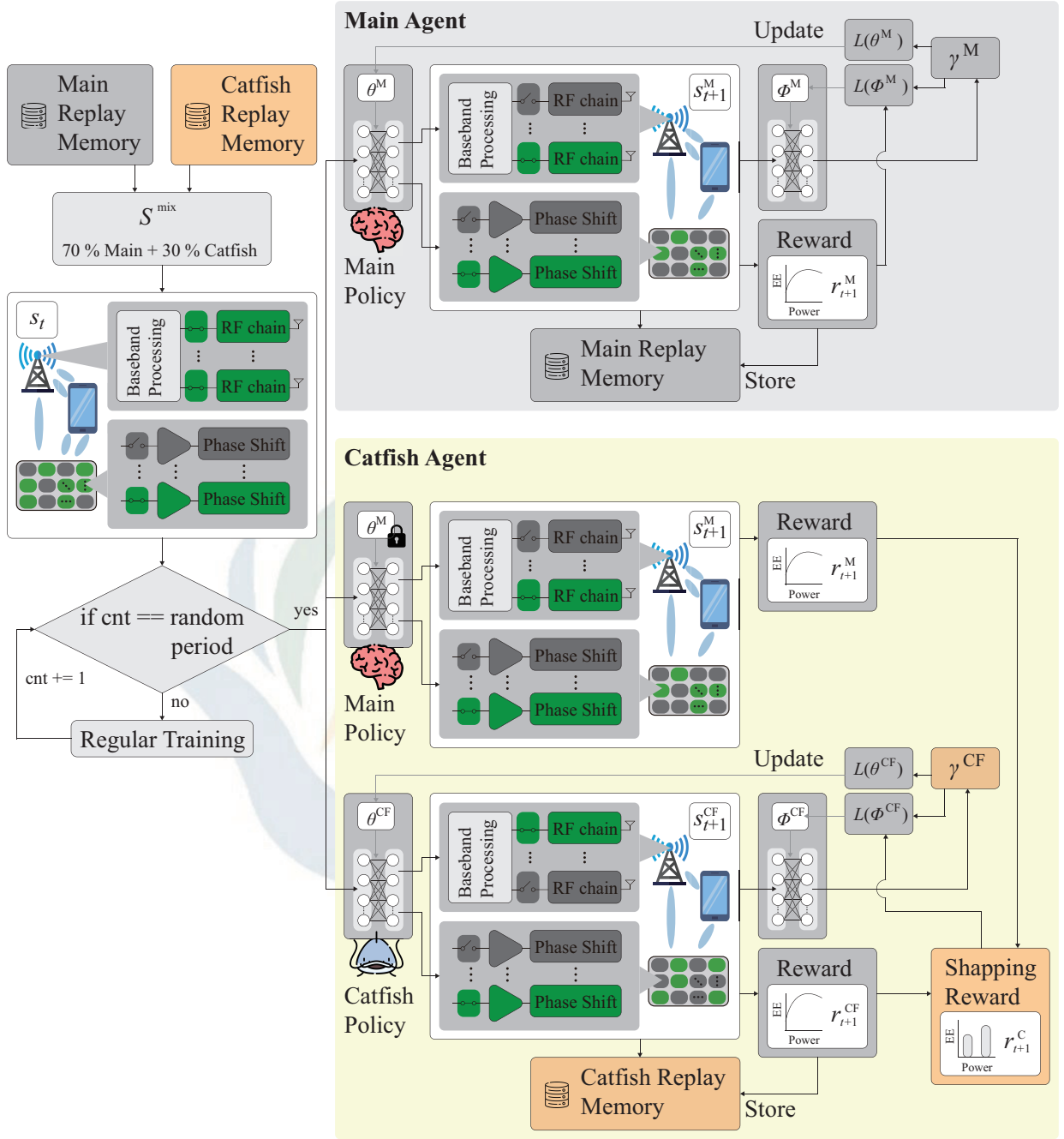


Figure 4.6: CDRL randomized periodic intervention training phase.

longer-term strategic cues. This study designs two training modes for CDRL. As shown in Fig. 4.5, in the normal training mode, the main agent and catfish agent each have independent experience replay memories, discount factors, and reward mechanisms, training independently without interference. As shown in Fig. 4.6, after a random period, the system enters the

intervention training mode. In this mode, a portion of the high EE experiences collected by the catfish agent is sampled and mixed into the main agent’s training batches. The main agent performs standard policy updates on the entire mixed batch as usual. However, since the batch contains high EE state-action pairs, experience-driven knowledge transfer is achieved. In the mixed batch $S^{\text{mix}} = S^M \cup S^{CF}$, 70% of the samples are drawn from the main replay memory S^M , and 30% from the catfish replay memory S^{CF} , where $S^M \in D^M, S^{CF} \in D^{CF}$. By orchestrating the above three components and the two-mode schedule, the catfish agent persistently supplies high-value trajectories while safeguarding the main policy from premature convergence.

A. Catfish Strategy 1: Experience Stratification

The purpose of experience stratification is to establish an elite challenger role by positioning the catfish agent as the “ignition point for high-level experiences”, thereby attracting other users or agents to challenge upward. This action provides different training data to the main agent and the catfish agent, assigning distinct tasks at each experience level. This strategy guides the main agent to adjust its policy by observing the catfish agent’s behavior. It enables both agents to better exploit their respective discount-factor characteristics during learning. Experiences with higher EE are stored in the catfish replay memory, while those with moderate EE are stored in the main replay memory, allowing the catfish agent to focus on “high-quality experiences” during training. The experience stratification flowchart is shown in Fig. 4.7 We package the randomly generated communication scenarios into datasets and use EE as the label to determine their storage locations. The threshold EE^{high} is set as the top 20% of the overall pre-generated data in phase 4.1, while EE^{min} is set as the top 50% performance of the pre-generated data.

- If $EE \geq EE^{\text{high}}$, the dataset is stored in the catfish replay memory as high-value data for the catfish agent. This set of experiences is denoted as $D^{CF} = \{D_1^{CF}, \dots, D_n^{CF}, \dots, D_{N/5}^{CF}\}$, where $1 \leq n \leq N/5$.

- Datasets with $EE^{\text{high}} \geq EE \geq EE^{\text{min}}$ are stored as typical communication states in the agent replay memory. This set of experiences is denoted as $D^M = \{D_1^M, \dots, D_n^M, \dots, D_{N/2}^M\}$, where $1 \leq n \leq N/2$.
- Datasets with $EE^{\text{min}} \geq EE$ are discarded to ensure that the two memories contain data with corresponding characteristics.

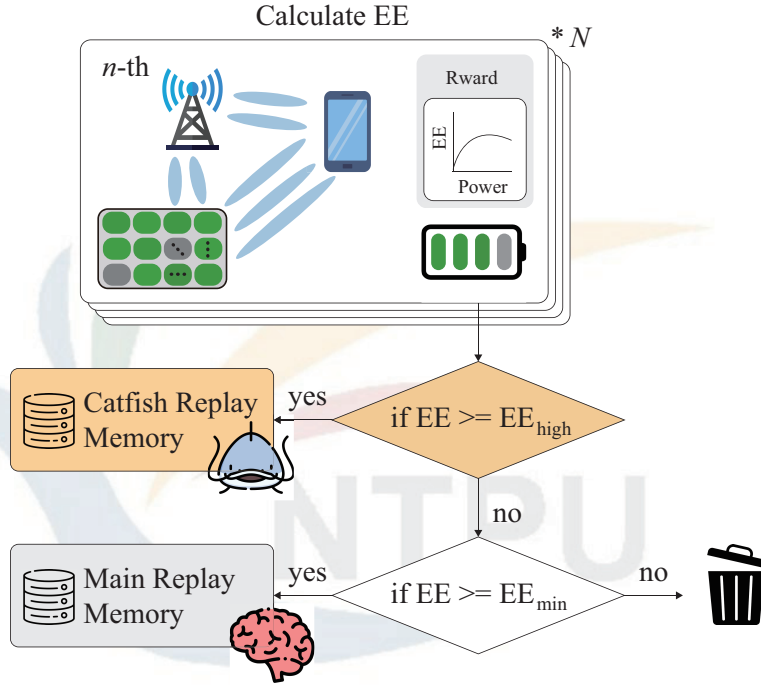


Figure 4.7: The flowchart of experience stratification.

B. Catfish Strategy 2: Asymmetric Discounting Strategy

The discount factor is a critical hyperparameter in reinforcement learning that determines an RL agent’s emphasis on future rewards. This hyperparameter influences the agent’s policy, convergence speed, and long-term optimization capability. In A. Catfish Strategy 1: Experience Stratification, this thesis describes why experiences are stratified. Our purpose in adjusting the discount factor is to extend the idea of assigning different tasks to different tiers, configuring the discount factors of the main agent and the catfish agent according to their respective tasks. This approach helps avoid strategy homogenization, thereby encouraging

differentiated strategies between agents. As a result, the main agent focuses on short-term demands to accelerate convergence, while the catfish agent emphasizes long-term objectives to pursue optimal outcomes.

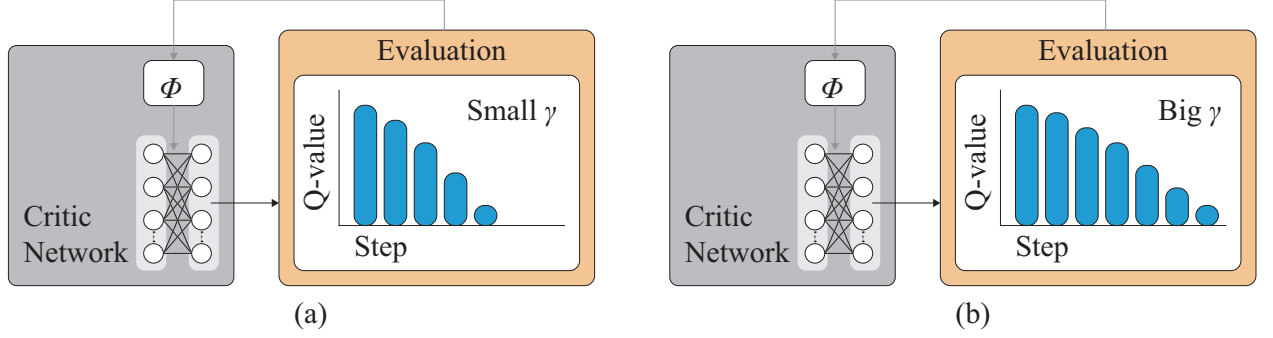


Figure 4.8: (a) Illustration of a smaller discount factor, and
(b) Illustration of a larger discount factor.

Observing equation (4.4), the discount factor causes the weight of the Q-value to decay exponentially as $Q_{\Phi}(s_{t+1}, \pi_{\theta}(s_{t+1}))$ unfolds into the future. The total return is given by $r_t^{\text{acc}} = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \gamma^3 r_{t+3} + \dots$. Thus, the closer the discount factor is to 1, the slower the decay, and the greater the emphasis on future rewards, as shown in Fig. 4.8(a) and Fig. 4.8(b). As shown in Fig. 4.8(a), the discount factor for the main agent is set to γ^M , while that for the catfish agent is set to γ^{CF} . According to Hou *et al.* [23], when an RL agent approaches the final outcome, a larger discount factor is beneficial for achieving better results. Accordingly, based on the prior experience stratification, superior experiences are assigned a larger discount factor for the catfish agent's tasks, while ordinary experiences receive a smaller discount factor for the main agent's tasks. Therefore, we set $\gamma^M \leq \gamma^{CF}$, allowing the catfish agent to learn more farsighted and superior policies, while the main agent is relatively myopic compared to the catfish agent.

C. Catfish Strategy 3: Adaptive Competitive Reward Mechanism

To further enhance the catfish agent's own capabilities in order to fulfill its role as an elite-level challenger. We modified the catfish agent's reward mechanism by incorporating an adaptive competitive reward mechanism (ACRM). By leveraging advanced heuristic methods

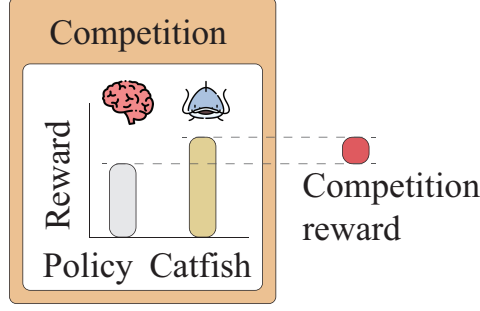


Figure 4.9: Illustration of adaptive competitive reward mechanism.

to guide the agent in extracting meaningful value from learning experiences, the stability of DRL convergence can be improved. The ACRM is similar to the Self-Adaptive Success Rate Sampling (SASR) proposed by Shen *et al.* [24]. This thesis expresses the ACRM's augmented reward function as:

$$r^C = r + \eta r^S \quad (4.8)$$

Here, r denotes the original EE reward from the environment, r^S represents the additional shaped reward, and η is the coefficient controlling the weight of this reward. When η approaches 0, the model degrades to the original reward model. As shown in Fig. 4.9, The additional shaped reward r^S at time t is expressed as:

$$r_t^S = r_t^{CF} - r_t^M \quad (4.9)$$

Here, r_t^M denotes the reward (energy efficiency) achieved by the main agent in the passive/active RIS communication system, while r_t^{CF} denotes the reward achieved by the catfish agent in the same system. The greater the lead of the catfish agent over the main agent on the same task, the larger the additional reward received by the catfish agent. Conversely, the catfish agent is penalized more when trailing behind, reinforcing the competitive dynamics.

Referring to equations (4.4)–(4.7), the update formula for the catfish agent with the adaptive competitive reward mechanism is defined. The temporal difference (TD) error δ^{CF} of the catfish agent is defined as:

Algorithm 1: The CDRL Training.

```
1 Input: Replay buffers  $\mathcal{D}^M$  (general),  $\mathcal{D}^{CF}$  (high-EE) from Phase I; actors  $\theta^M, \theta^{CF}$ ;
   critics  $\Phi^M, \Phi^{CF}$ ; discounts  $\gamma^M, \gamma^{CF}$ .
2 for  $e = 0, 1, 2, \dots$  do
3   Reset env; get  $s_0$  (channels:  $Q, h_k, g_k$ , actions:  $w_k^{BB}, F^{RF}, C^{RF}, \Lambda^{RIS}, \phi^{RIS},$ 
    $B^{RIS}/A^{RIS}$ , power terms).
4   for  $t = 0, 1, 2, \dots$  do
5      $a_t^M \leftarrow \pi_{\theta^M}(s_t^M); a_t^{CF} \leftarrow \pi_{\theta^{CF}}(s_t^{CF})$ .
6     Execute  $a_t^M$ ; observe  $(r_t^M, s_{t+1}^M)$ . Execute  $a_t^{CF}$ ; observe  $(r_t^{CF}, s_{t+1}^{CF})$ .
7     Compute  $EE_t^M, EE_t^{CF}$ .
8     Add to Replay buffers  $\mathcal{D}^M$ ; Add to Replay buffers  $\mathcal{D}^{CF}$ .
9     // Main updates
10    Sample  $\mathcal{B}^M = (s, a, r, s')1^B \sim \mathcal{D}_M$ .
11     $\delta_t = r_t + \gamma Q_{\Phi}(s_{t+1}, \pi_{\theta}(s_{t+1})) - Q_{\Phi}(s_t, a_t)$ .
12     $\Phi_{t+1} = \Phi_t - \mu_c \nabla_{\Phi} L(\Phi_t)$ .
13     $\theta_{t+1} = \theta_t - \mu_a \left[ \nabla_a Q_{\Phi}(s_t, a) \Big|_{a=\pi_{\theta}(s_t)} \right] \nabla_{\theta} \pi_{\theta}(s_t)$ .
14    // Catfish updates with ACRM
15     $r_t^S \leftarrow r_t^{CF} - r_t^M$ ; set  $r_t^C \leftarrow r_t^{CF} + \eta r_t^S$ .
16    Sample  $\mathcal{B}^{CF} = (s, a, r^C, s')1^B \sim \mathcal{D}^{CF}$ .
17     $\delta_t^{CF} = r_t^{CF} + \eta r_t^S + \gamma^{CF} Q_{\Phi^{CF}}^{CF}(s_{t+1}, \pi_{\theta^{CF}}^{CF}(s_{t+1})) - Q_{\Phi^{CF}}^{CF}(s_t, a_t^{CF})$ .
18     $\Phi_{t+1}^{CF} = \Phi_t^{CF} - \mu_c^{CF} \nabla_{\Phi^{CF}} L(\Phi_t^{CF})$ .
19     $\theta_{t+1}^{CF} = \theta_t^{CF} - \mu_a^{CF} \left[ \nabla_a Q_{\Phi^{CF}}^{CF}(s_t, a) \Big|_{a=\pi_{\theta^{CF}}^{CF}(s_t)} \right] \nabla_{\theta^{CF}} \pi_{\theta^{CF}}^{CF}(s_t)$ .
20   end
21 end
22 Output: Trained  $\theta^M, \Phi^M, \theta^{CF}, \Phi^{CF}$ .
```

$$\delta_t^{CF} = r_t^{CF} + \eta r_t^S + \gamma^{CF} Q_{\Phi^{CF}}^{CF}(s_{t+1}, \pi_{\theta^{CF}}^{CF}(s_{t+1})) - Q_{\Phi^{CF}}^{CF}(s_t, a_t^{CF}) \quad (4.10)$$

Here, $\pi_{\theta^{CF}}^{CF}(s_{t+1})$ denotes the action given by the catfish actor at the next state. γ^{CF} is the discount factor of the catfish agent, determining the importance of the future Q-value $Q_{\Phi^{CF}}^{CF}(s_{t+1}, \pi_{\theta^{CF}}^{CF}(s_{t+1}))$.

The catfish critic uses a neural network to approximate the Q-value function $Q_{\Phi^{CF}}^{CF}(s_t, a_t^{CF})$ and updates the network parameters by minimizing the mean squared error (MSE) loss. The loss function is defined as:

$$L(\Phi^{CF}) = (\delta_t^{CF})^2 \quad (4.11)$$

Then, gradient descent is used to update the critic's parameters Φ^{CF} as:

$$\Phi_{t+1}^{CF} = \Phi_t^{CF} - \mu_c^{CF} \nabla_{\Phi^{CF}} L(\Phi_t^{CF}) \quad (4.12)$$

Here, μ_c^{CF} is the learning rate, and $\nabla_{\Phi^{CF}}$ denotes the gradient with respect to Φ^{CF} of the loss function $L(\Phi_t^{CF})$. The parameter update formula for the catfish actor θ^{CF} is:

$$\theta_{t+1}^{CF} = \theta_t^{CF} - \mu_a^{CF} \left[\nabla_a Q_{\Phi^{CF}}^{CF}(s_t, a) \Big|_{a=\pi_{\theta^{CF}}^{CF}(s_t)} \right] \nabla_{\theta^{CF}} \pi_{\theta^{CF}}^{CF}(s_t) \quad (4.13)$$

Here, μ_a^{CF} is the learning rate.

4.3 Catfish-effect-based DRL optimization of energy efficiency

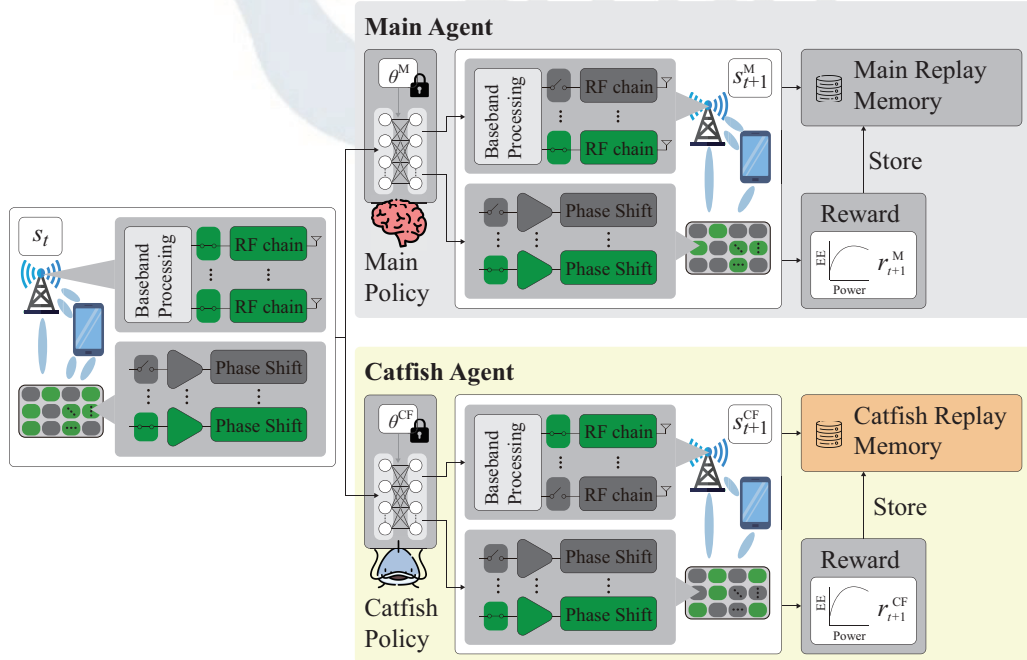


Figure 4.10: The flowchart of trained CDRL.

In this phase, we describe how the trained CDRL is executed. In the preceding phase 4.2, we optimized the main agent's weights Φ^M and θ^M as well as the catfish agent's weights Φ^{CF} and θ^{CF} , whereby stimulation from the catfish agent enabled CDRL to acquire the optimal policy. As shown in Fig. 4.10, during this phase the main agent performs only state-input to action-output inference and no longer executes any gradient updates—i.e., only the main actor is required—and the catfish agent performs only performance monitoring—i.e., only the catfish actor is required. We use a different set of states in this phase than during training while maintaining the same input-output parameters. As in the previous stage, the experiment begins by configuring the required parameters according to the experimental specifications: the number of transmit antennas N_t , the number of RF chains N_{RF} , the number of RIS elements N_{RIS} , the number of single-antenna UEs N_{UE} , the maximum transmit power $P_{BS}^{\max}$ and the RIS position. Next, following the experimental rules, we generate random spatial layouts for the BS, UE and RIS and compute the BS–RIS, RIS–UE and BS–UE channel matrices Q , h_k and g_k . We then initialize the baseband digital beamformer w_k^{BB} , the RF analog beamformer F^{RF} , the RF-chain switch matrix C^{RF} , the active RIS power-amplification matrix Λ^{RIS} , the phase-shift matrix ϕ^{RIS} , the passive RIS element-switch matrix B^{RIS} and the active RIS element-switch matrix A^{RIS} , and treat these initial parameters as the initial state $[s_0]$ handed to the trained CDRL for subsequent inference. The CDRL algorithm we proposed is shown in Algorithm 1.

Chapter 5

Experimental Result

In this chapter, the experimental setup is explained in section 5.1 and the performance evaluation is discussed in section 5.2, the ablation studies are presented in section 5.3, and the RIS simulator is introduced in section 5.4.

5.1 Experimental Setup

This section details the environment setup used for training the proposed model. The experiments are conducted on a system running Windows 10 as the underlying operating system. The programming environment is built using Python 3.8.16, with NumPy 1.26.4 and PyTorch 2.3.1 to support the implementation of reinforcement learning algorithms. All training and evaluation processes are performed using an NVIDIA GeForce RTX 2080 Ti GPU to ensure computational efficiency. The downlink communication system environment is configured in accordance with [10], with detailed parameters listed in Table 5.1.

5.2 Performance analysis

This section presents a performance comparison between the proposed approach and its rivals. The experiments are divided into four scenarios based on active/passive RIS and the presence or absence of element switching. We use the energy efficiency and sum rate defined in chapter 3 as performance metrics. These performance metrics are:

Table 5.1: Simulation environment and system parameters.

Symbol	Description	Value
ψ_{BS}	BS coordinates	(0, 0) m
ψ_{RIS}	RIS coordinates	(50, 10) m
ψ_{UE}	UE coordinates	random on circle: (50, 0), $r = 3$ m
N_t	BS antennas	8
N_{RF}	RF chains at BS	6
N_{RIS}	RIS reflecting elements	6
N_{UE}	Number of UEs	2
$P_{\text{BS}}^{\text{max}}$	BS transmit-power range	[10, 20] dBm
R^{min}	Minimum spectral efficiency	1 bit/s/Hz
σ^2	Noise power	−70 dBm
δ^2	Interference power (threshold)	−80 dBm
P_C	Circuit power	−10 dBm
P_{DC}	DC power	−5 dBm
P^{static}	Static power	10 dBm

- Energy efficiency is an overall system performance indicator and is defined as the ratio of sum rate to total energy consumption, evaluating the transmission capability per unit energy.
- Sum rate is the transmission speed metric, representing the total amount of data received by all users per unit time, used to assess the capacity of multi-user channel resources.

To assess the energy-efficiency performance of the proposed reinforcement learning model, we benchmark it against three methods, including BPOF. In addition to schemes with adjustable element switching, configurations with all RIS elements fully activated are included as reference benchmarks. Fully activated means that all reflecting elements on the RIS cannot be switched off. This aims to assess the impact of RIS element switching on the energy efficiency and sum rate of the communication system. The experiments are further divided into active and passive RIS to evaluate the effect of active RIS amplifiers on system energy efficiency and sum rate. The proposed CDRL scheme is represented by a red line with solid diamonds. The

fully activated RIS scheme is shown as a red line with solid circles. The three baselines are as follows:

- The first is a basic actor-critic reinforcement learning model [11] (black line with hollow squares) serving as the baseline, with the fully activated RIS variant depicted as a black line with hollow triangles.
- The second is the BPOF mathematical analytical solution [10] (blue line with solid diamonds), based on the same communication system architecture, with the fully activated RIS variant shown as a blue line with solid circles.
- The third baseline is the teacher-student learning framework [21] (green line with hollow squares), serving as the primary competitor, with the fully activated RIS variant represented by a green line with hollow triangles. This comparison aims to evaluate differences in knowledge transfer mechanisms within dual-agent architectures.

5.2.1 Energy Efficiency

“Energy efficiency” vs. “transmission power consumption” experimental results for active/passive RIS are plotted in Fig. 5.1(a)/ 5.3(a). According to the results, as the transmission power constraint is relaxed, the energy efficiency of all algorithms improves. The reason is that the reduced energy constraint increases schedulable elements, resulting in a higher total transmission rate. Observing Fig. 5.1(a), for the adjustable-switch active RIS scheme of this thesis’s CDRL, the best EE performance is achieved owing to the adjustable switches and active RIS, together with the catfish agent’s effective impetus for the main agent to explore strategies, thereby learning superior beamforming and component-configuration combinations. The all-on-switch active RIS scheme of this thesis’s CDRL exhibits lower EE performance because when all elements are fully on, excessive element power consumption reduces overall energy efficiency. Observing Fig. 5.3(a), for the adjustable-switch passive RIS scheme of this thesis’s CDRL, the best EE performance among all passive schemes is achieved. The all-on-switch

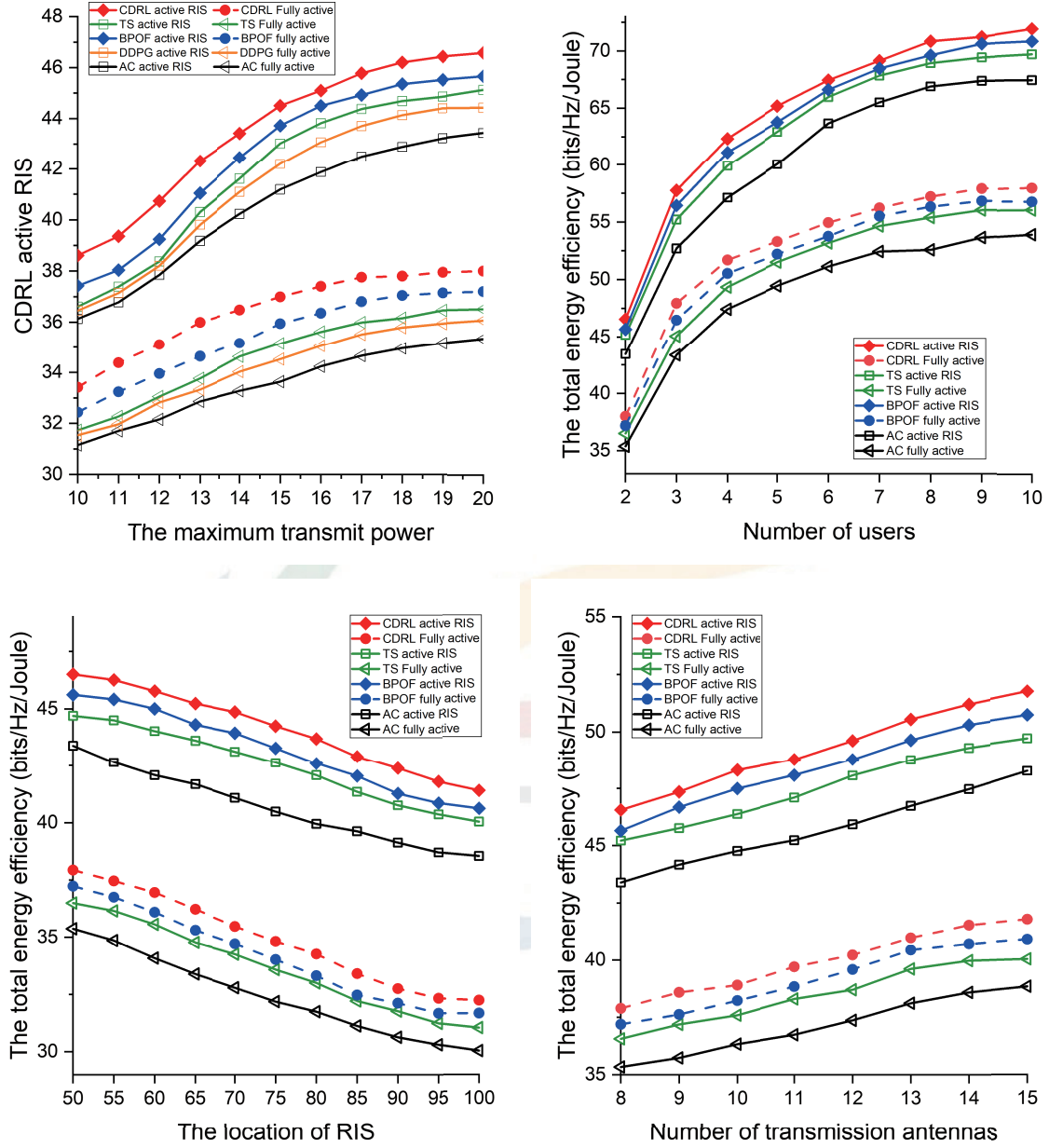


Figure 5.1: The performance comparison of active RIS in terms of energy efficiency.

passive RIS scheme of this thesis's CDRL exhibits lower EE performance, and because optimization constraints such as the minimum-rate requirement must be satisfied, EE differences among passive RIS schemes are small. Compared with existing methods, the proposed CDRL framework demonstrates higher EE performance under various transmission power constraints.

“Energy efficiency” vs. “number of users” experimental results for active/passive RIS

are plotted in Fig. 5.1(b)/ 5.3(b). According to the results, as the user number increases, the energy efficiency of all algorithms improves. The reason is that an increased number of users served by the BS yields a higher total transmission rate. Observing Fig. 5.1(b), for the adjustable-switch active RIS scheme of this thesis's CDRL, the best EE performance is achieved owing to the adjustable switches and active RIS. The all-on-switch active RIS scheme of this thesis's CDRL exhibits lower EE performance due to excessive element power consumption reducing overall energy efficiency. Observing Fig. 5.3(b), for the adjustable-switch passive RIS scheme of this thesis's CDRL, the best EE performance among all passive schemes is achieved. The all-on-switch passive RIS scheme of this thesis's CDRL exhibits lower EE performance. Compared with existing methods, the proposed CDRL framework demonstrates higher EE performance under various user numbers. To investigate both the overall energy efficiency and the average user energy efficiency, the results of the active RIS experiment are presented in Fig. 5.2. According to the results, while the overall energy efficiency improves, the average user energy efficiency inevitably decreases. This decline may be attributed to the mutual interference among users during signal reception.

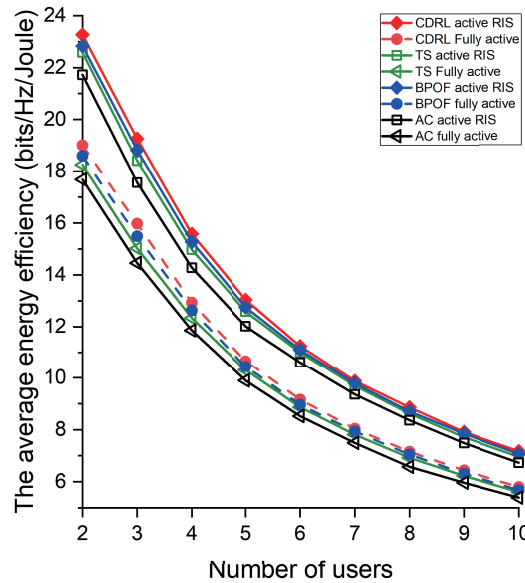


Figure 5.2: The performance comparison of active RIS in terms of energy efficiency.

“Energy efficiency” vs. “the location of RIS” experimental results for active/passive RIS are plotted in Fig 5.1(c)/ 5.3(c). The results indicate a decline in EE across all algorithms as the RIS deployment is positioned increasingly farther from the reference point (50, 10). The reason is that the farther the RIS is from the UE, the weaker the channel gain from the BS to the UE, resulting in a lower total transmission rate. Observing Fig. 5.1(c), for the adjustable-switch active RIS scheme of this thesis’s CDRL, the best EE performance is achieved owing to the adjustable switches and active RIS. The all-on-switch active RIS scheme of this thesis’s CDRL exhibits lower EE performance due to excessive element power consumption reducing overall energy efficiency. Observing Fig. 5.3(c), for the adjustable-switch passive RIS scheme of this thesis’s CDRL, the best EE performance among all passive schemes is achieved. The all-on-switch passive RIS scheme of this thesis’s CDRL exhibits lower EE performance. Compared with existing methods, this thesis’s CDRL framework demonstrates higher energy efficiency performance under various RIS deployment positions.

“Energy efficiency” vs. “number of transmission antennas” experimental results for active/passive RIS are plotted in Fig. 5.1(d)/ 5.3(d). According to the results, as the number of transmit antennas increases, the energy efficiency of all algorithms improves. The reason is that an increased antenna number leads to higher transmission signal strength, resulting in a higher total transmission rate. Observing Fig. 5.1(d), for the adjustable-switch active RIS scheme of this thesis’s CDRL, the best EE performance is achieved owing to the adjustable switches and active RIS. The all-on-switch active RIS scheme of this thesis’s CDRL exhibits lower EE performance due to excessive element power consumption reducing overall energy efficiency. Observing Fig. 5.3(d), for the adjustable-switch passive RIS scheme of this thesis’s CDRL, the best EE performance among all passive schemes is achieved. The all-on-switch passive RIS scheme of this thesis’s CDRL exhibits lower EE performance. Compared with existing methods, this thesis’s CDRL framework demonstrates higher EE performance under various numbers of transmit antennas.

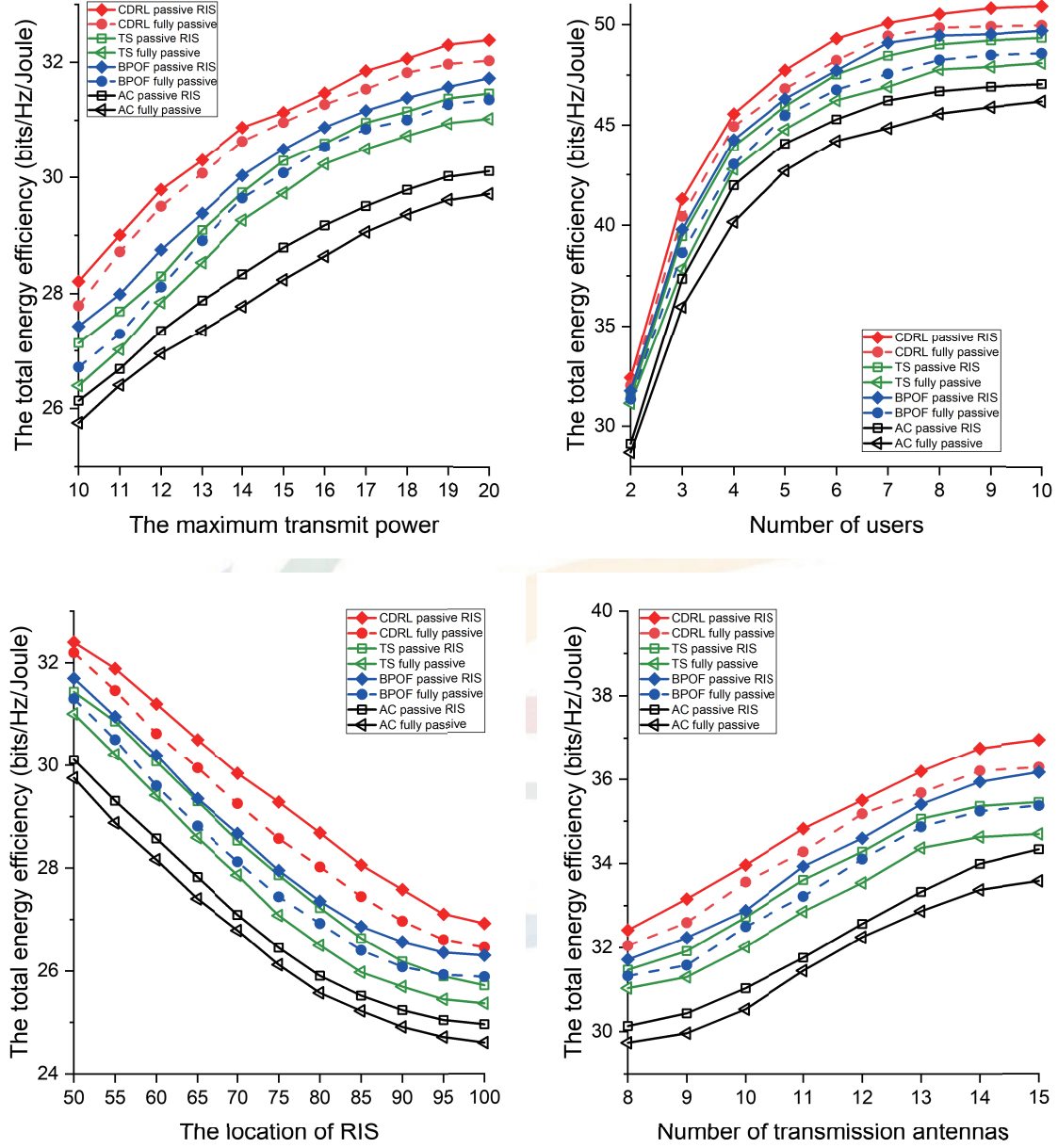


Figure 5.3: The performance comparison of passive RIS in terms of energy efficiency.

5.2.2 Sum Rate

“Sum rate” vs. “transmission power consumption” experimental results for active/passive RIS are plotted in Fig. 5.4(a)/ 5.6(a). According to the results, as the transmission power constraint is relaxed, the spectral efficiency of all algorithms improves because fewer energy restrictions increase the number of schedulable elements, thus raising the total transmission rate.

Observing Fig. 5.4(a), for this thesis's CDRL adjustable-switch active RIS scheme, the best SE performance is achieved owing to the combined benefits of adjustable switches, active RIS and the catfish agent's effective incentive of the main agent to explore new strategies, thereby learning superior beam-forming and component-configuration combinations. The all-on-switch active RIS scheme of this thesis's CDRL exhibits lower SE performance because when all elements are fully on, redundant elements increase channel noise, causing a slight SE degradation. Observing Fig. 5.6(a), for this thesis's CDRL adjustable-switch passive RIS scheme, the best SE performance among all passive schemes is achieved. The all-on-switch passive RIS scheme of this thesis's CDRL exhibits lower SE performance, and because optimization constraints such as the minimum-rate requirement must be met, SE differences among passive RIS schemes remain small. Compared with existing methods, this thesis's CDRL framework demonstrates higher transmission efficiency under various transmission power constraints.

"Sum rate" vs. "number of users" experimental results for active/passive RIS are plotted in Fig. 5.4(b)/ 5.6(b). According to the results, as the user number increases, the spectral efficiency of all algorithms improves because a larger number of users served by the BS yields a higher total transmission rate. Observing Fig. 5.4(b), for this thesis's CDRL adjustable-switch active RIS scheme, the best SE performance is achieved owing to adjustable switches and active RIS. The all-on-switch active RIS scheme of this thesis's CDRL exhibits lower SE performance because redundant elements increase channel noise, causing a slight SE degradation. Observing Fig. 5.6(b), for this thesis's CDRL adjustable-switch passive RIS scheme, the best SE performance among all passive schemes is achieved. The all-on-switch passive RIS scheme of this thesis's CDRL exhibits lower SE performance. Compared with existing methods, this thesis's CDRL framework demonstrates higher transmission efficiency under various user numbers. To investigate both the overall transmission rate and the average user transmission rate, the results of the active RIS experiment are illustrated in Fig. 5.5. The results indicate that although the overall transmission rate increases, the average user transmission rate decreases. This degradation may be caused by mutual interference among

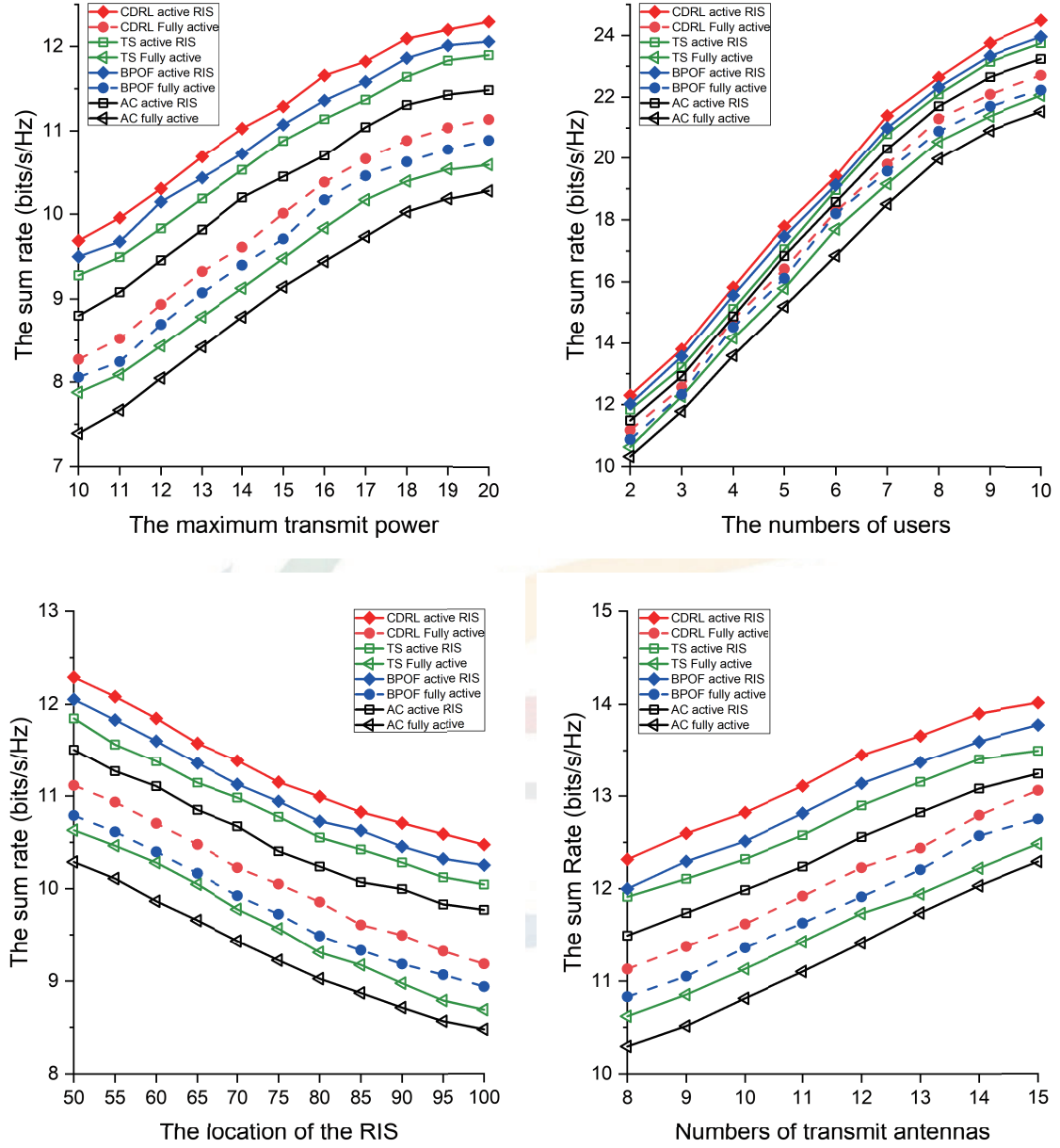


Figure 5.4: The performance comparison of active RIS in terms of sum rate.

users during signal reception.

“Sum rate” vs. “the location of RIS” experimental results for active/passive RIS are plotted in Fig. 5.4(c)/ 5.6(c). The results indicate a decline in SE across all algorithms as the RIS deployment is positioned increasingly farther from the reference point (50, 10). According to the results, as the RIS deployment position moves away from (50, 0), the spec-

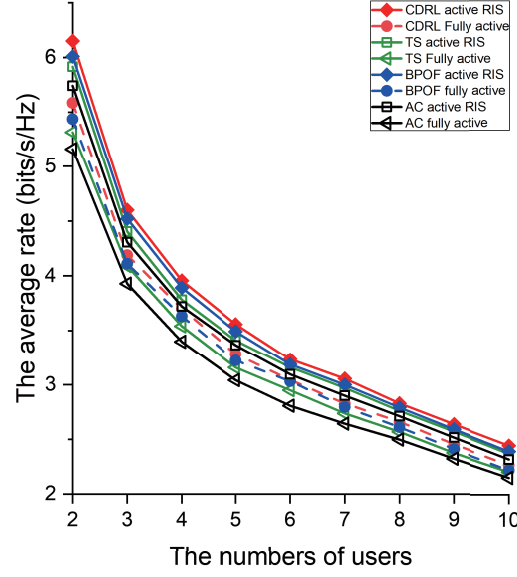


Figure 5.5: The performance comparison of active RIS in terms of sum rate.

tral efficiency of all algorithms decreases because the farther the RIS is from the UE, the weaker the channel gain from the BS to the UE, resulting in a lower total transmission rate. Observing Fig. 5.4(c), for this thesis's CDRL adjustable-switch active RIS scheme, the best SE performance is achieved owing to adjustable switches and active RIS. The all-on-switch active RIS scheme of this thesis's CDRL exhibits lower SE performance because redundant elements increase channel noise, causing a slight SE degradation. Observing Fig. 5.6(c), for this thesis's CDRL adjustable-switch passive RIS scheme, the best SE performance among all passive schemes is achieved. The all-on-switch passive RIS scheme of this thesis's CDRL exhibits lower SE performance. Compared with existing methods, this thesis's CDRL framework demonstrates higher transmission efficiency under various RIS deployment positions.

“Sum rate” vs. “number of transmission antennas” experimental results for active/passive RIS are plotted in Fig. 5.4(d)/ 5.6(d). According to the results, as the number of transmit antennas increases, the spectral efficiency of all algorithms improves because more antennas boost the transmission signal, increasing the total transmission rate. Observing Fig. 5.4(d), for this thesis's CDRL adjustable-switch active RIS scheme, the best

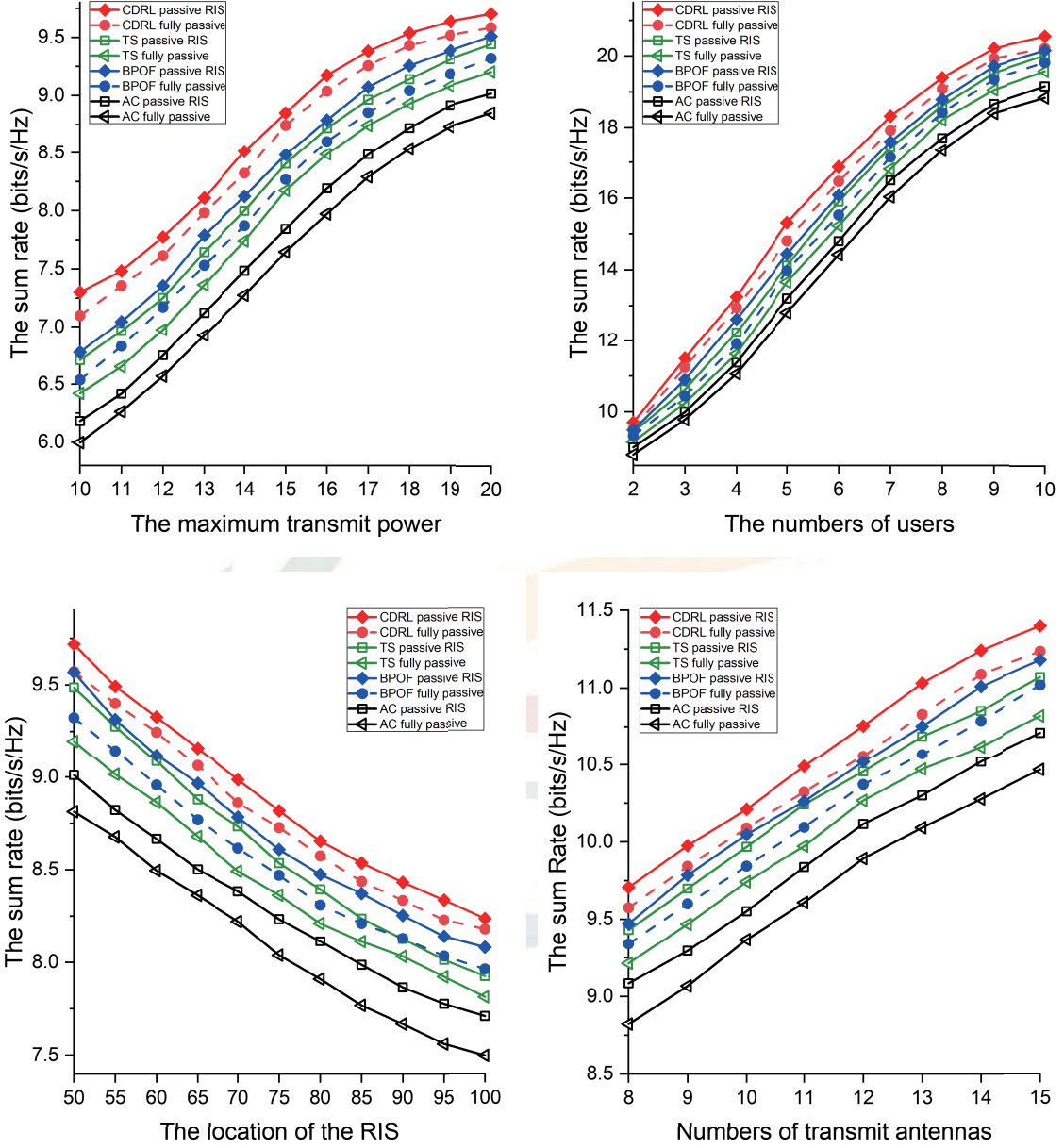


Figure 5.6: The performance comparison of passive RIS in terms of sum rate.

SE performance is achieved owing to adjustable switches and active RIS. The all-on-switch active RIS scheme of this thesis's CDRL exhibits lower SE performance because redundant elements increase channel noise, causing a slight SE degradation. Observing Fig. 5.6(d), for this thesis's CDRL adjustable-switch passive RIS scheme, the best SE performance among all passive schemes is achieved. The all-on-switch passive RIS scheme of this thesis's CDRL ex-

hibits lower SE performance. Compared with existing methods, this thesis’s CDRL framework demonstrates higher transmission efficiency under various numbers of transmit antennas.

5.3 Ablation Experiment

This thesis conducts an ablation experiment to elucidate the contributions of its three catfish strategies to CDRL. In each experiment, one strategy is removed from the complete CDRL scheme (red line with solid diamonds). The first ablation removes Catfish Strategy 1: Experience Stratification (orange line with solid circles) and replaces it with an unsegmented experience-replay buffer. The second ablation removes Catfish Strategy 2: Asymmetric Discounting Strategy (blue line with solid circles), instead using an identical discount factor ($\gamma^M = \gamma^{CF} = 0.95$). The third ablation removes Catfish Strategy 3: Adaptive Competitive Reward Mechanism (green line with solid circles), so the catfish agent receives no extra reward. The fourth one uses only Catfish Strategy 1: Experience Stratification (orange line with solid diamond markers), where the experience replay memory is replaced with classified experiences. The fifth one uses only Catfish Strategy 2: Asymmetric Discounting Strategy (blue line with solid diamond markers), where different discount factors are applied ($\gamma^M = 0.95, \gamma^{CF} = 0.99$). The sixth one uses only Catfish Strategy 3: Adaptive Competitive Reward Mechanism (green line with solid diamond markers), where the catfish agent applies only the additional reward.

“Transmission rate vs. transmission power constraint” results for the active RIS ablation are plotted in Fig. 5.7. The unsegmented replay yields a moderate performance drop, likely because all experiences, regardless of reward magnitude, remain equally probable in sampling. The identical discount-factor ablation causes a larger degradation, suggesting that similar objectives for catfish and main agents reduce the main agent’s exploratory stimulus. Omitting the adaptive competitive reward mechanism has the smallest effect, implying that the baseline reward density already sufficiently drives CDRL training. Using only the pre-classified experience replay memory yields a moderate improvement, possibly due to the lack of support from the other two strategies. Using only different discount factors achieves the

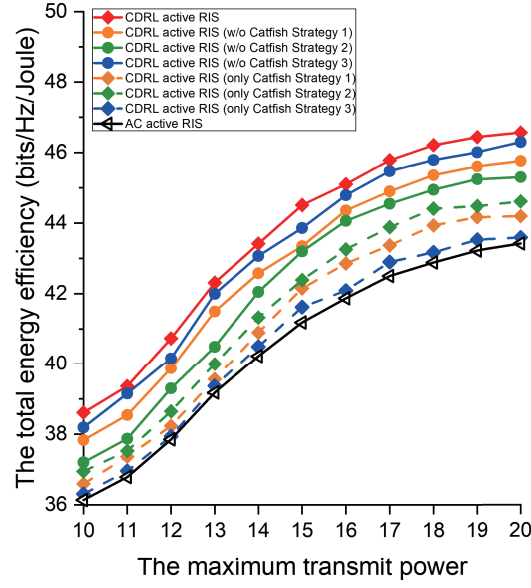


Figure 5.7: The ablation experiment of active RIS in terms of energy efficiency.

greatest improvement, which may result from the effective division of tasks. Using only the catfish agent without the Adaptive Competitive Reward Mechanism shows a smaller improvement, possibly because the additional reward provides limited benefit.

5.4 The RIS Simulator

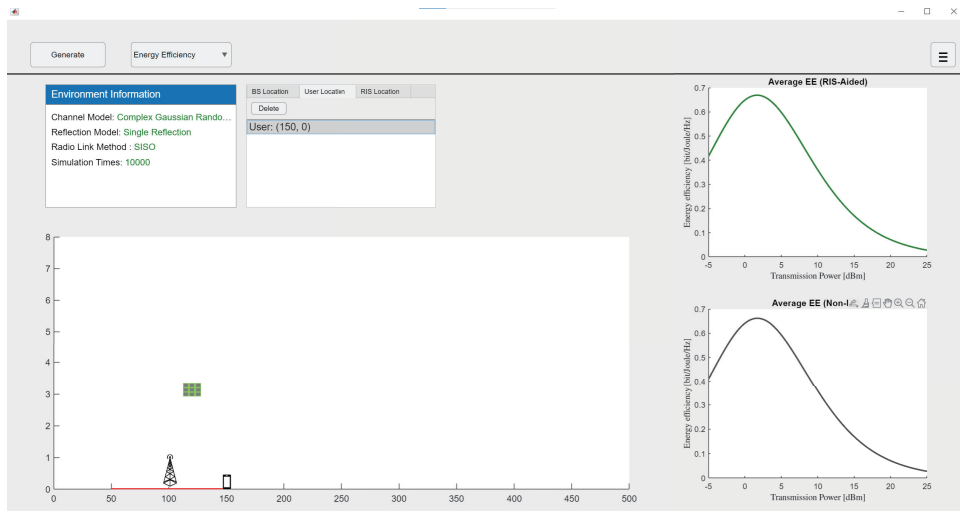


Figure 5.8: The RIS simulator.

We extended the improved BPOF [10] to develop a simulator for the RIS communication environment, as shown in Fig. 5.8. By configuring the positions of the BS, RIS, and UE, as well as adjusting detailed parameters of the relevant components, the simulator can evaluate the EE performance in this communication scenario. Moreover, additional functions such as RIS element switching were incorporated to enable energy-efficient joint on/off control of RF chains and RIS elements.



Chapter 6

Conclusion

This thesis proposes a Catfish-effect-based Deep Reinforcement Learning (CDRL) framework for energy-efficient integrated switching control of RF chains and RIS elements in an RIS-assisted 6G network. Consider a hybrid beamforming architecture at the base station that combines a baseband digital precoder with an RF analog beamformer. Each element on the communication devices can be switched independently. Three core designs for the catfish agent are proposed, all aimed at ensuring its policy performance surpasses the main agent. First, in the data pre-generation phase, all experiences are routed according to an EE threshold. Second, the catfish agent's discount factor is set larger to focus on long-term return estimation. Third, an adaptive competitive reward mechanism is implemented for the catfish agent, encouraging it to continually maintain its performance lead. Through the dual-agent design enables the Catfish Agent to guide the Main Agent toward better policies and avoid local optima. Simulation results show that CDRL outperforms baseline DRL, analytical methods, and teacher-student models under both active and passive RIS settings. Future work may explore extending the framework to more practical or large-scale scenarios.

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